

APOLLO 9

**LM SYSTEMS
EVALUATION CHECKLIST**

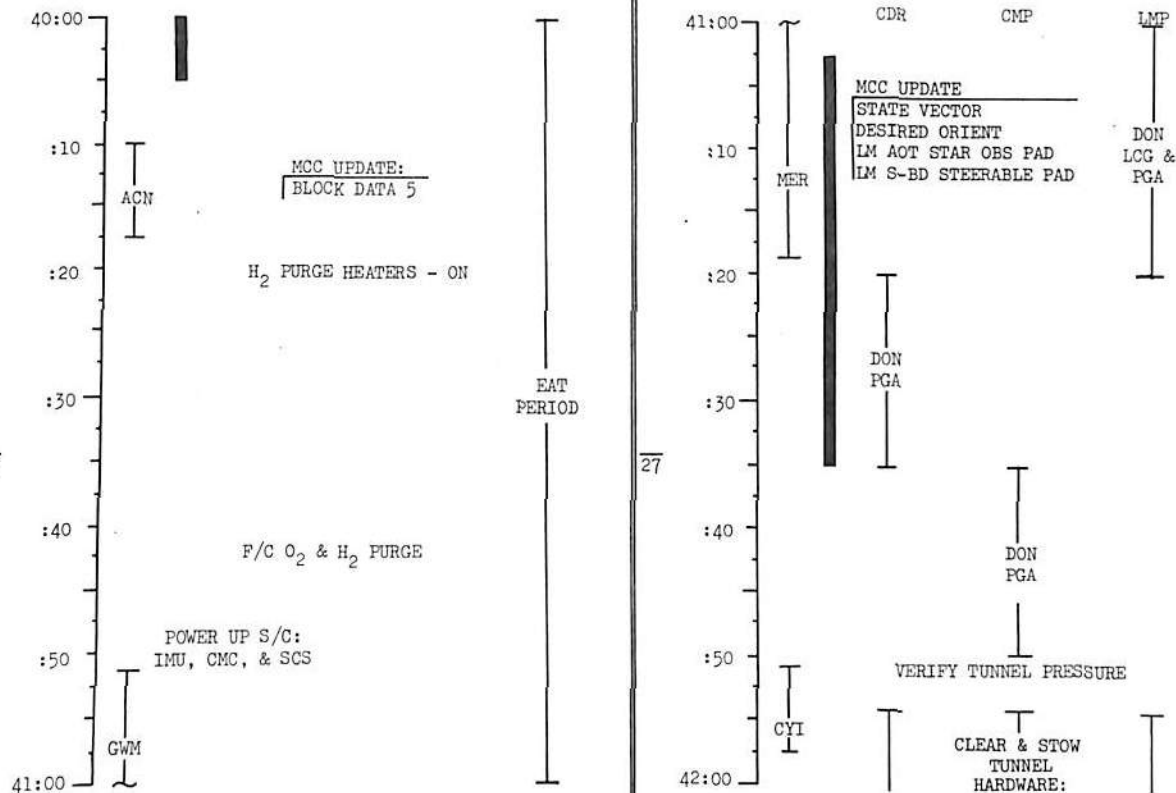
PART NO

S/N

SKB32100013-301

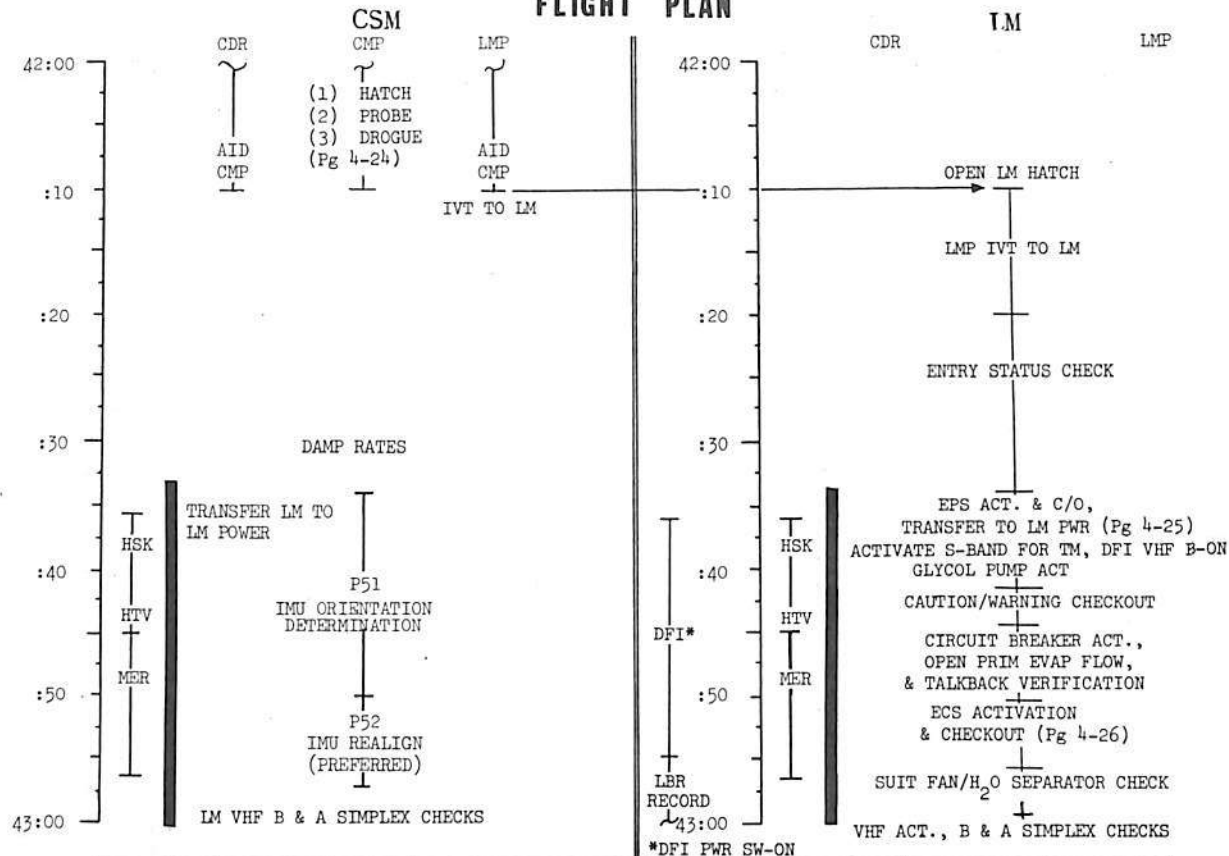
1004

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	40:00 - 42:00	3/26-27	3-23

FLIGHT PLAN

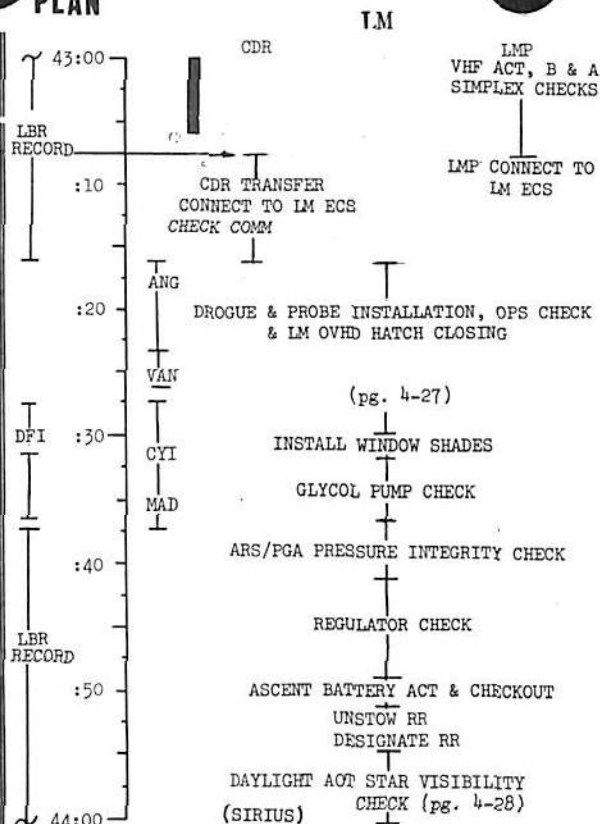
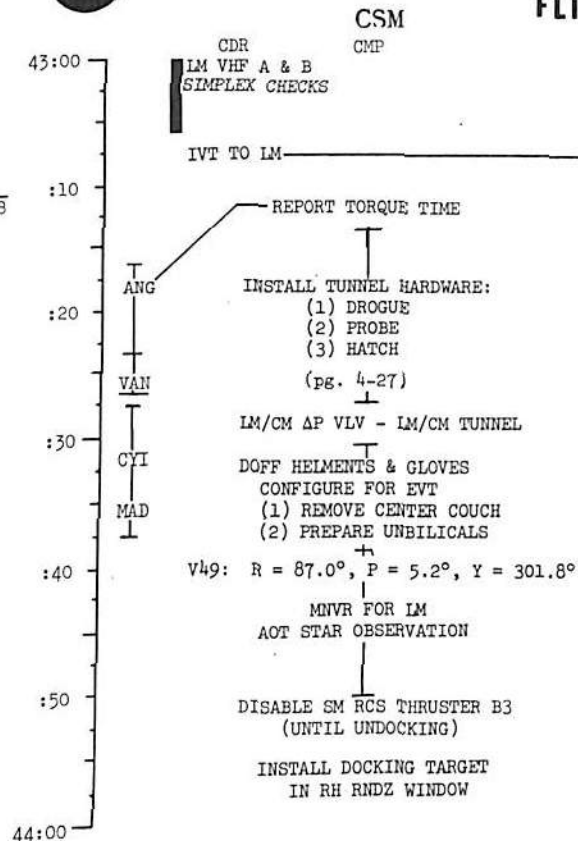


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	42:00 - 43:00	3/27	3-24

MSC FORM 1186 (SEP 67)

FLIGHT PLANNING BRANCH

FLIGHT PLAN



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	43:00 - 44:00	3/27-28	3-25

CSM
CMP

FLIGHT PLAN

CDR

IM

LMP

44:00
:10
:20
:30
:40
:50
45:00

CRO

HSK

MER

ANG

CO₂ FILTER CHANGE NO. 4
6 TO B, 4 TO B5

S-Band CONFERENCE
(MSFN RELAY) (pg 4-31)

44:00
:10
:20
:30
:40
:50
45:00

CRO

HSK

MER

ANG

LBR :40
RECORD

S-BAND STEERABLE
ANTENNA CHECK
(pg. 4 - 30)

S-BAND BACKUP
VOICE & KEYING CHECK
(pg. 4 - 31)

S-BAND CONFERENCE
(MSFN RELAY)
(pg. 4 - 34)

EXT. LTG CHECKS
& COAS LTG

SECONDARY S-BAND
CHECK (pg 4 - 31)

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	44:00 - 45:00	3/23-29	3-26

MSC FORM 1186 (SEP 67)

FLIGHT ENGINEERING BRANCH

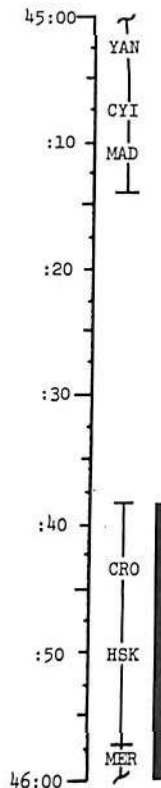
CSM
CMP

FLIGHT PLAN

LM

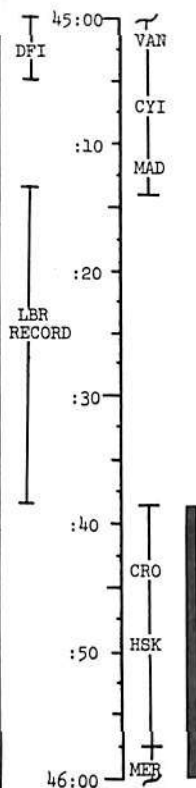
CDR

LMP



CSM 1-WAY RELAY
WITH LM SIMULATING PLSS

CSM 1-WAY RELAY
SIMULTANEOUS WITH
LM 1-WAY RELAY



LANDING GEAR DEPLOY (Pg 4-33)

PLSS PREP:

- (1) INSTALL ISA
- (2) UNSTOW TV
- (3) UNSTOW PLSS BATT
- (4) BLEED DOWN PLSS HOSES
- (5) UNSTOW OPS
- (6) VERIFY LIOH CARTRIDGE
- (7) ATTACH OPS TO PLSS
- (8) INSTLL PLSS BAT IN PLSS
- (9) UNSTOW RCU
- (10) CONNECT RCU TO PLSS
- (11) PREP FOOD

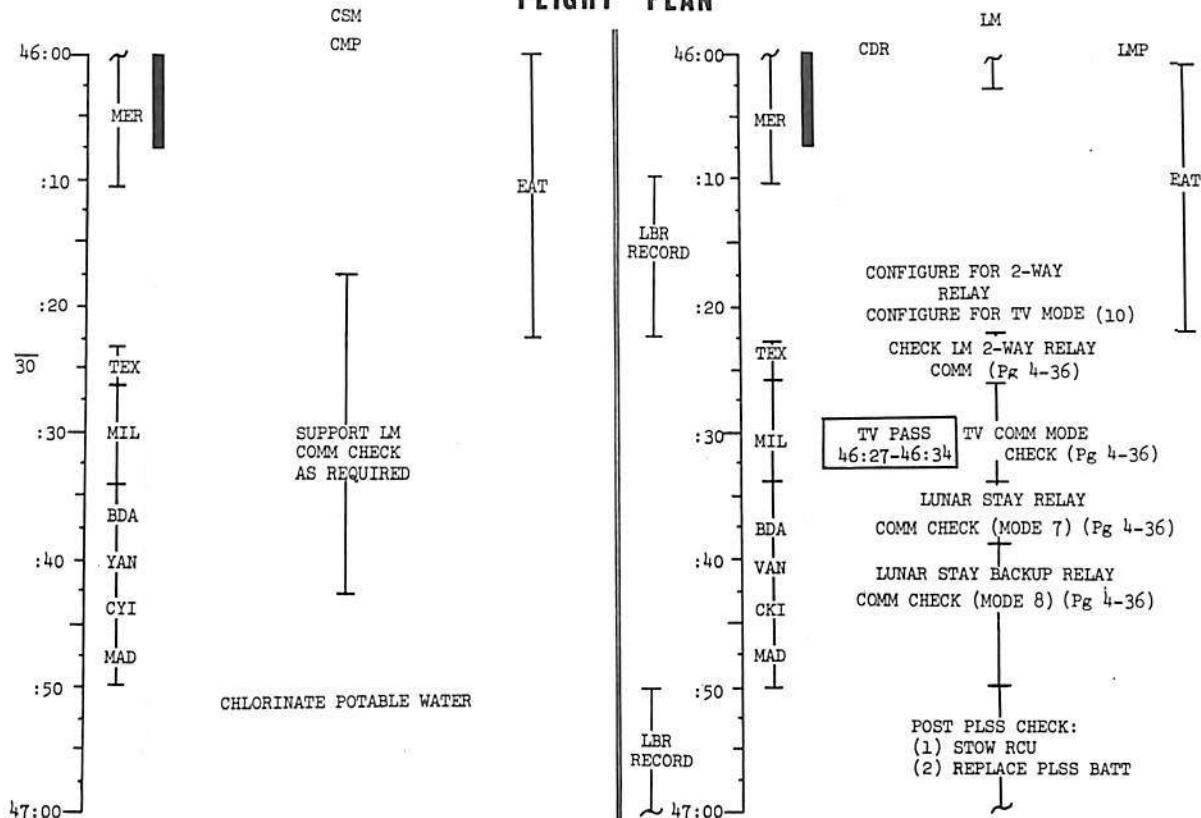
PLSS COMM CHECK WITH LM
PLSS POS 1 & 4

CSM 1-WAY RELAY
WITH LM SIMULATING PLSS
(PLSS IN POS 4)

LM 1-WAY RELAY
USING PLSS SIMULTANEOUS
WITH CSM 1-WAY RELAY

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	45:00 - 46:00	3/29	3-27

FLIGHT PLAN

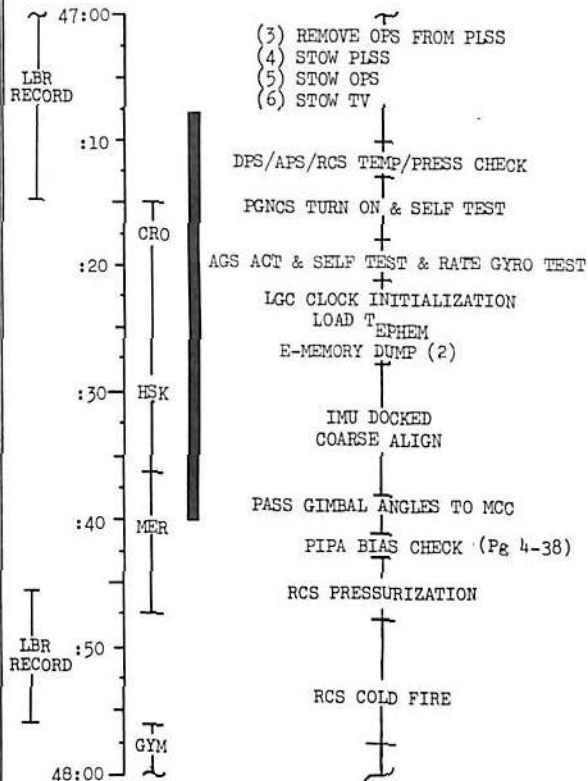
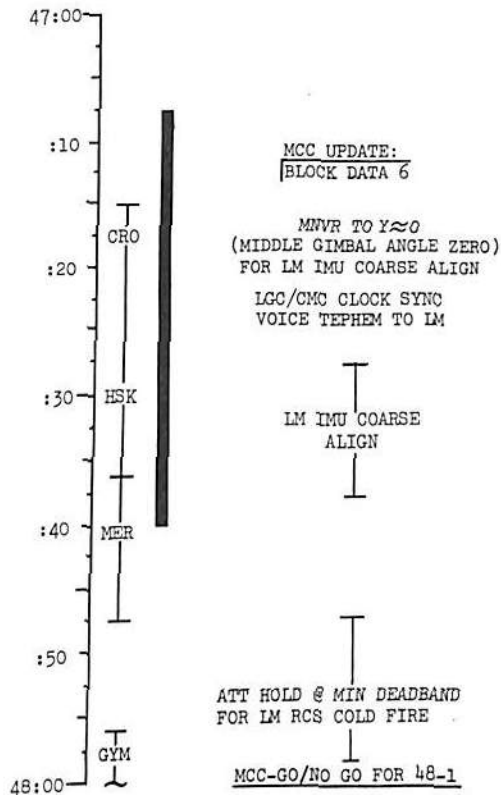


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEB 3, 1969	46:00 - 47:00	3/29-30	3-28

CSM
CMP

FLIGHT PLAN

CDR IM
LMP

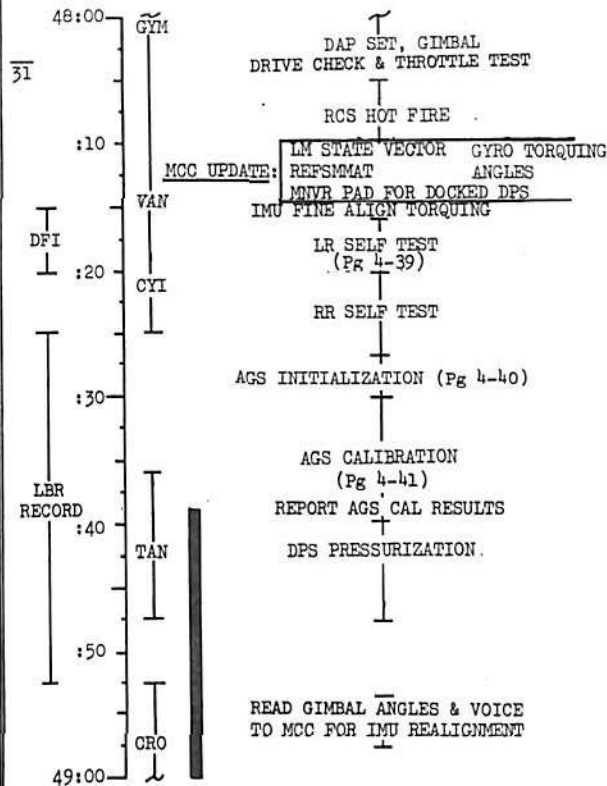
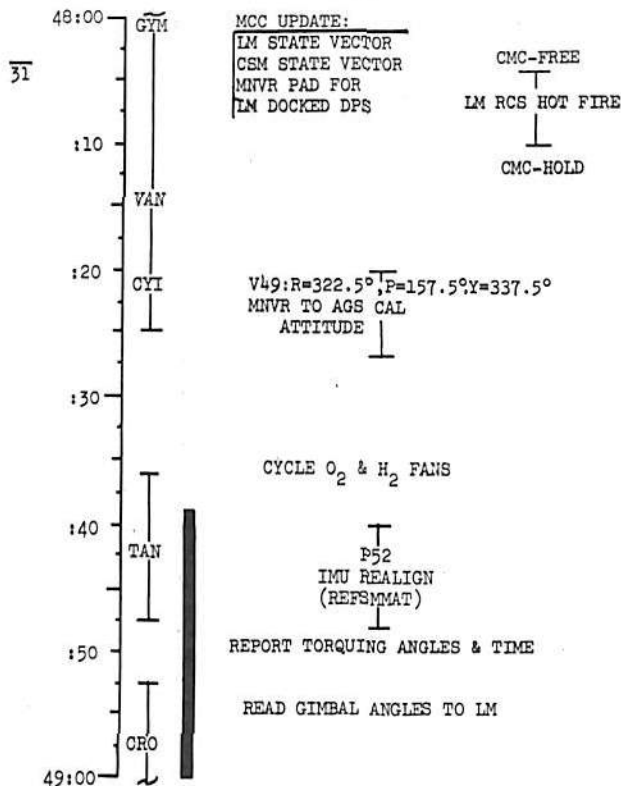


MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	47:00 - 48:00	3/30	3-29

CSM
CMP

FLIGHT PLAN

CDR IM LMP



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	48:00 - 49:00	3/30-31	3-30

MSC FORM 1186 (SEP 67)

FLIGHT ENGINEERING BRANCH

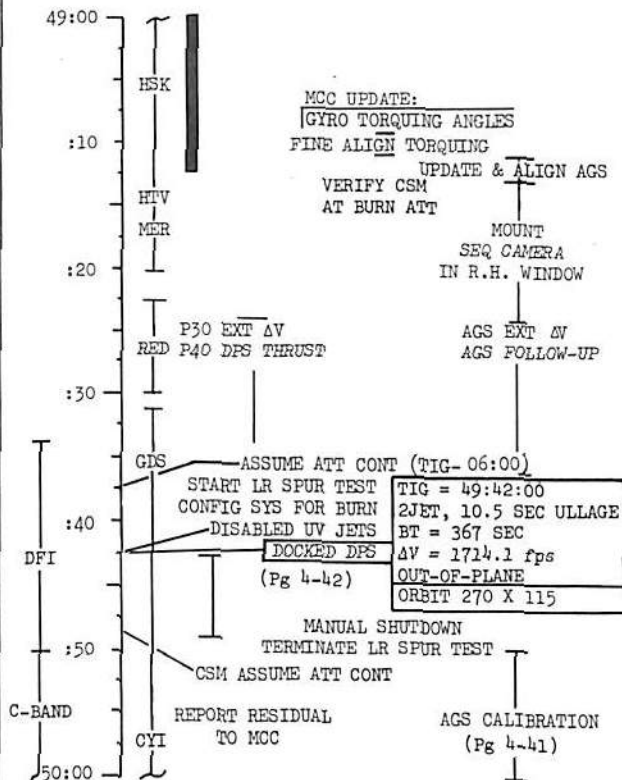
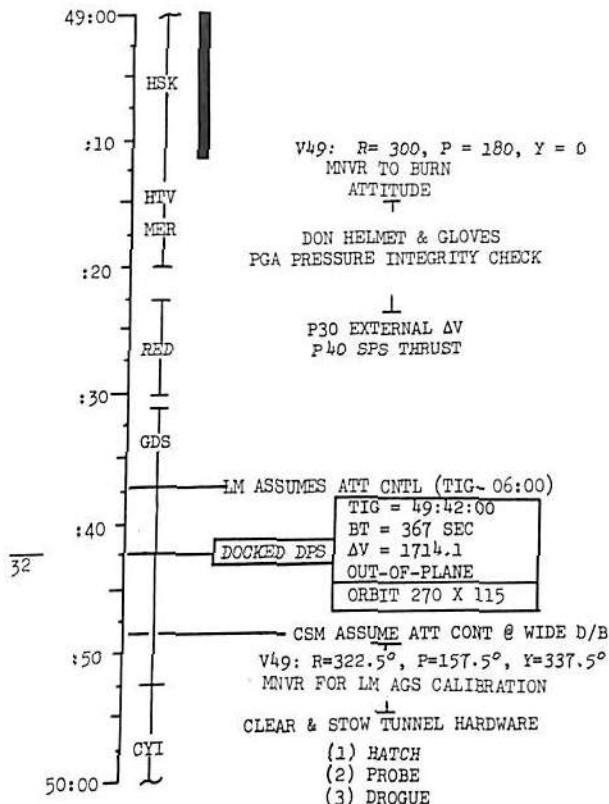
CSM
CMP

FLIGHT PLAN

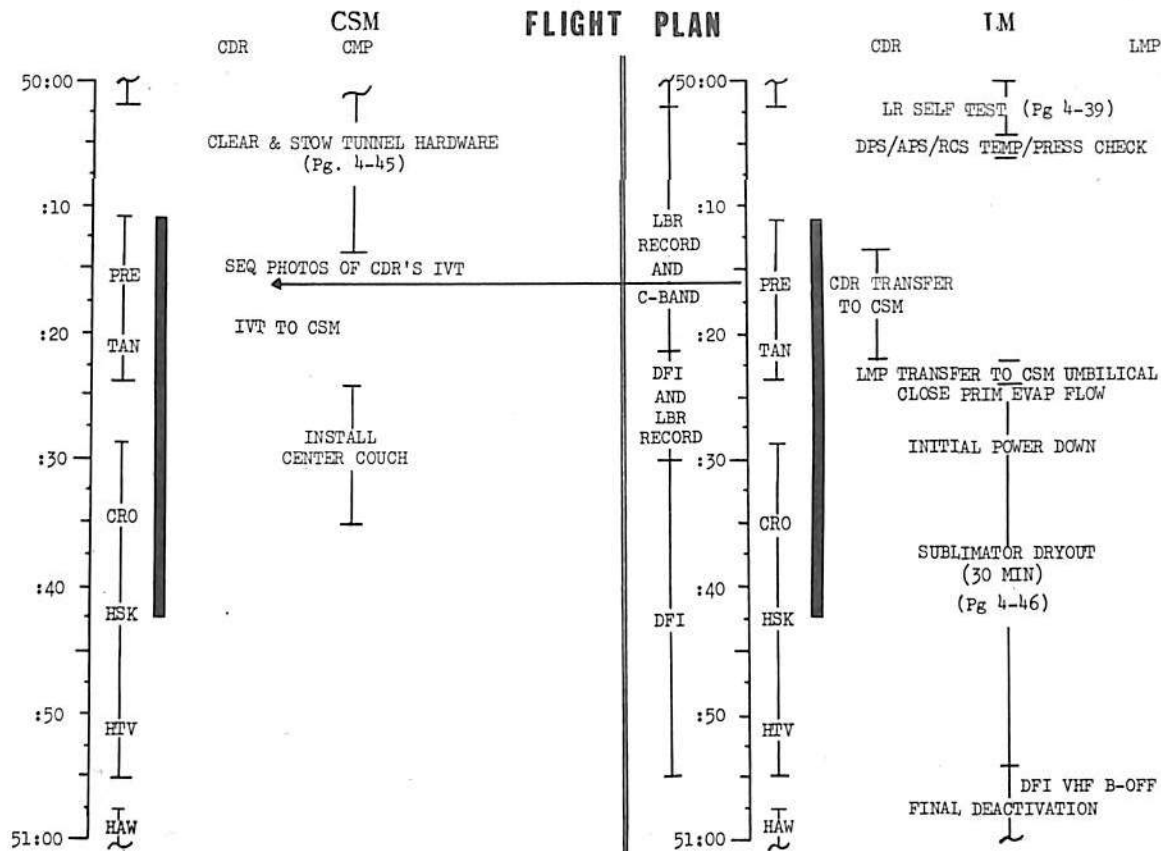
CDR

LM

LMP



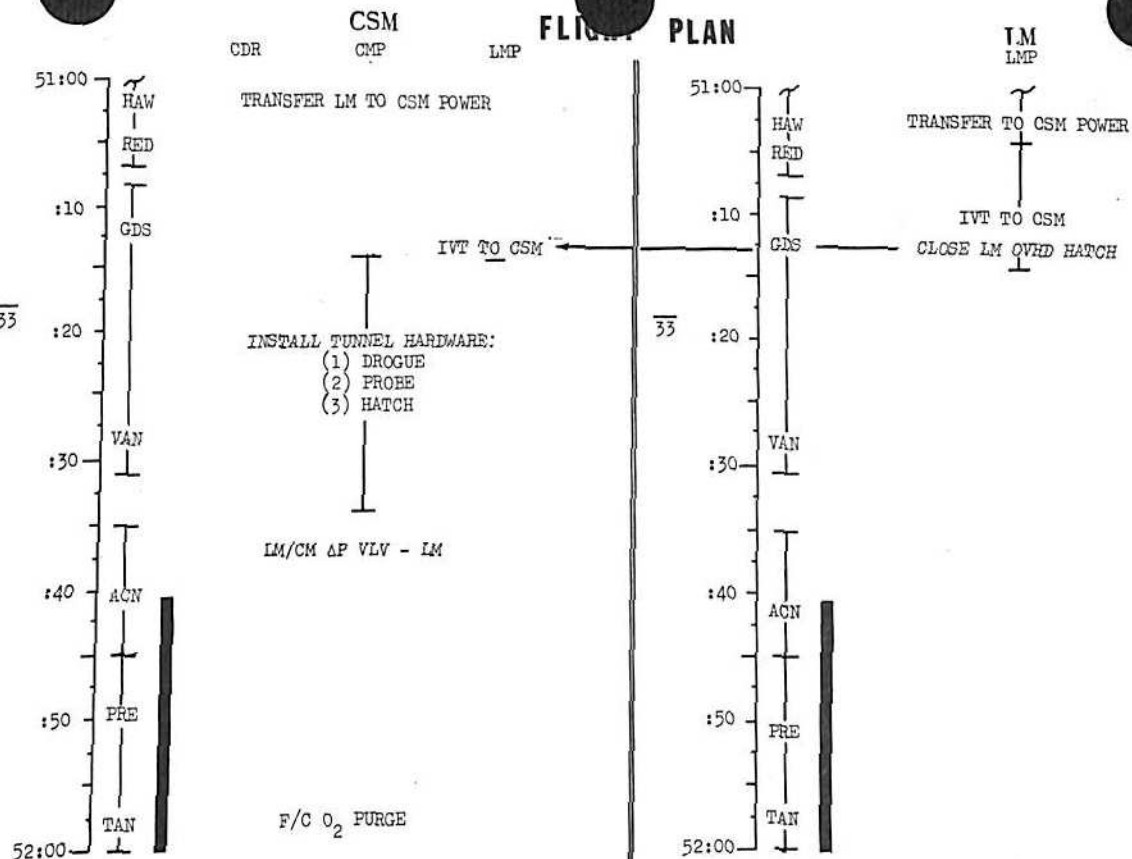
MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	49:00 TO 50:00	3/31-32	3-31



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	50:00 - 51:00	3/32	3-32

MSC FORM 1186 (SEP 67)

FLIGHT PLANNING BRANCH



MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	51:00 - 52:00	3/32-33	3-33

FLIGHT PLAN

CDR

CMP

LMP

CDR

CMP

LMP

52:00
:10
:20
:30
:40
:50
53:00

CRO

HTV

HAW

GDS

TEX

GBM

MCC UPDATE :

STATE VECTOR

TGT LOAD

MNVR PAD

53:00
:10
:20
:30
:40
:50
54:00

ANG

ACN

PRE

TAN

CRO

GWM

P30 EXTERNAL AV
P40 SPS THRUST

P52
IMU REALIGN
(PREFERRED)

LOAD DAP
P40 SPS THRUST

SXT STAR CHECK

CYCLE O₂ & H₂ FANS

MCC-SPS GO/NO GO

34

MISSION	EDITION	DATE	TIME	DAY/REV	PAGE
APOLLO 9	FINAL	FEBRUARY 3, 1969	52:00 - 54:00	3/33-34	3-34

MSC FORM 1186 (SEP 67)

FLIGHT PLANNING BRANCH

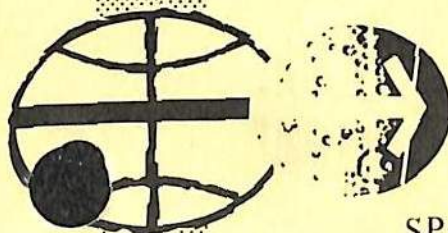


NATIONAL AERONAUTICS
AND SPACE ADMINISTRATION

APOLLO IX LM-3
FINAL
FLIGHT CREW
SYSTEMS
EVALUATION
CHECKLIST

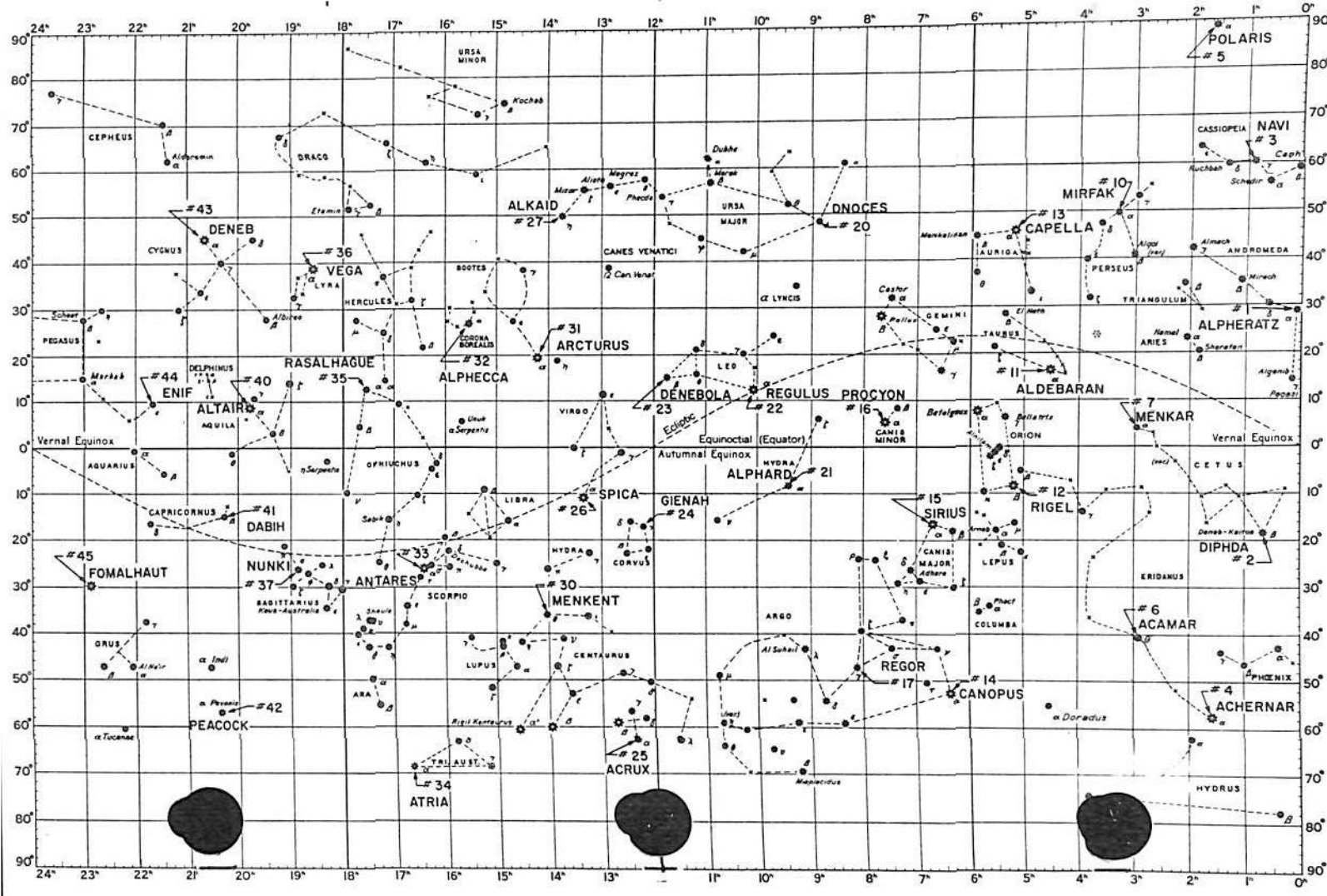
PREPARED BY
GUIDANCE & CONTROL
SECTION
FLIGHT CREW SUPPORT
DIVISION

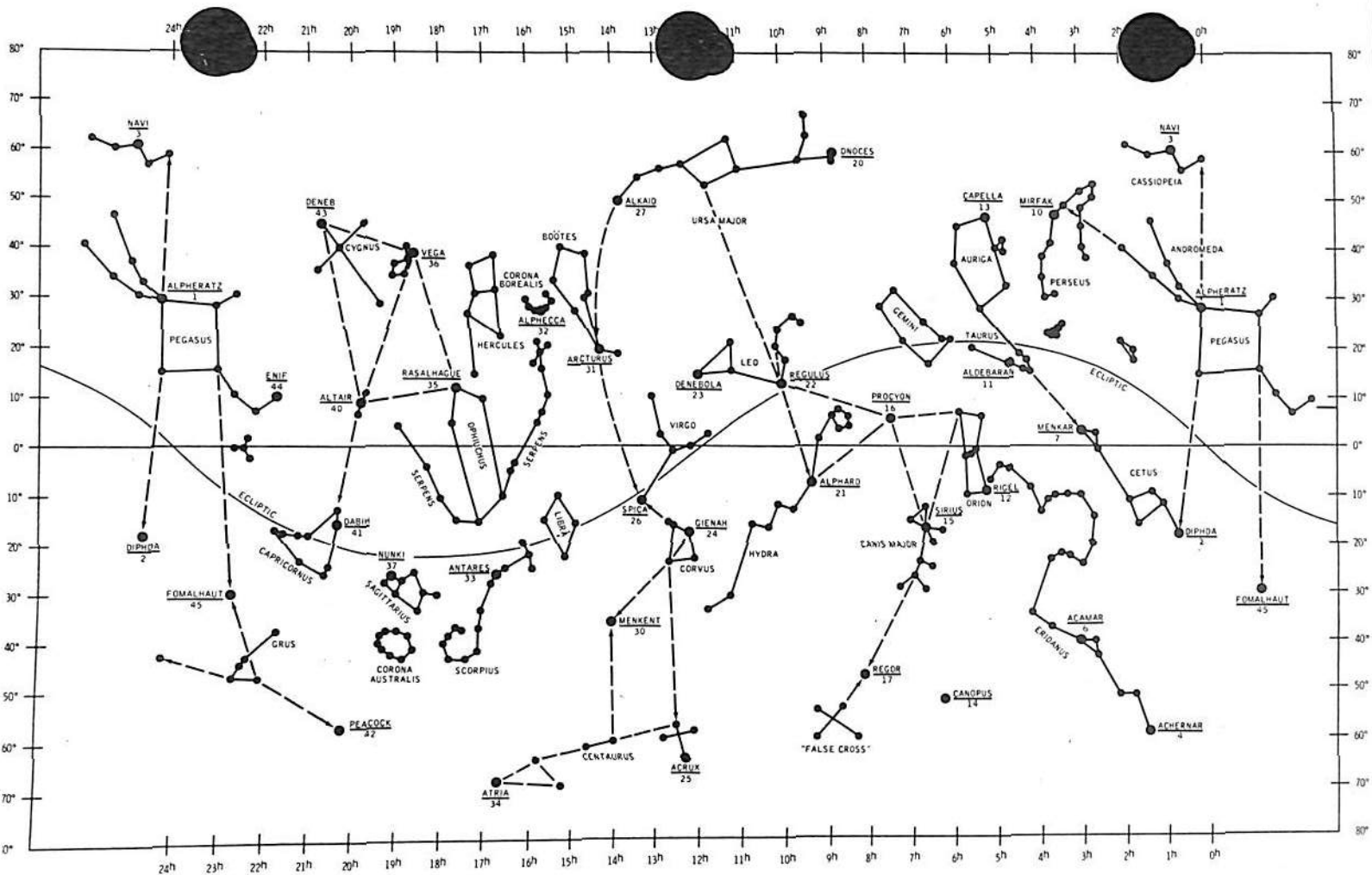
FEBRUARY 24, 1969

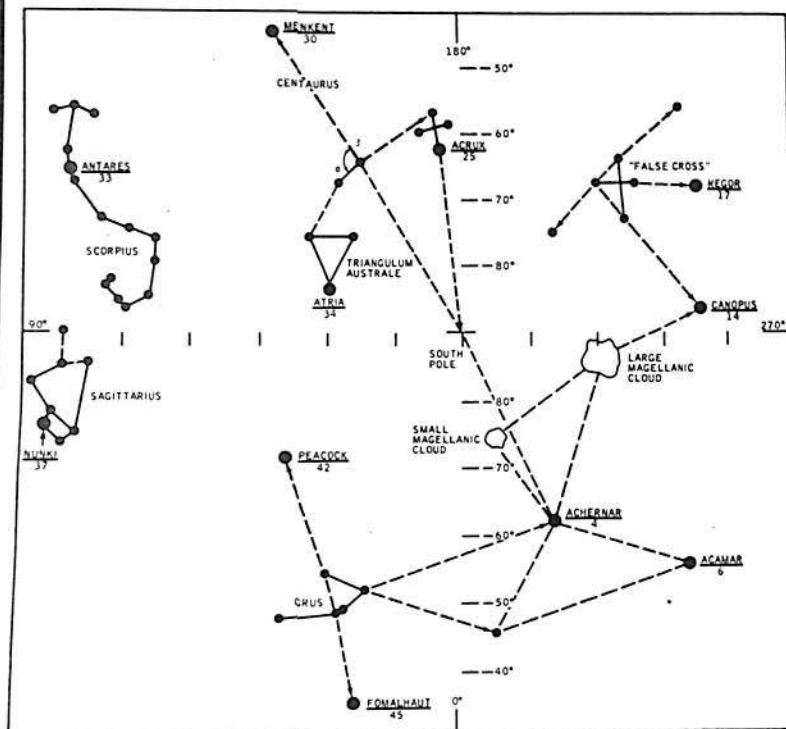


MANNED
SPACECRAFT CENTER
HOUSTON, TEXAS

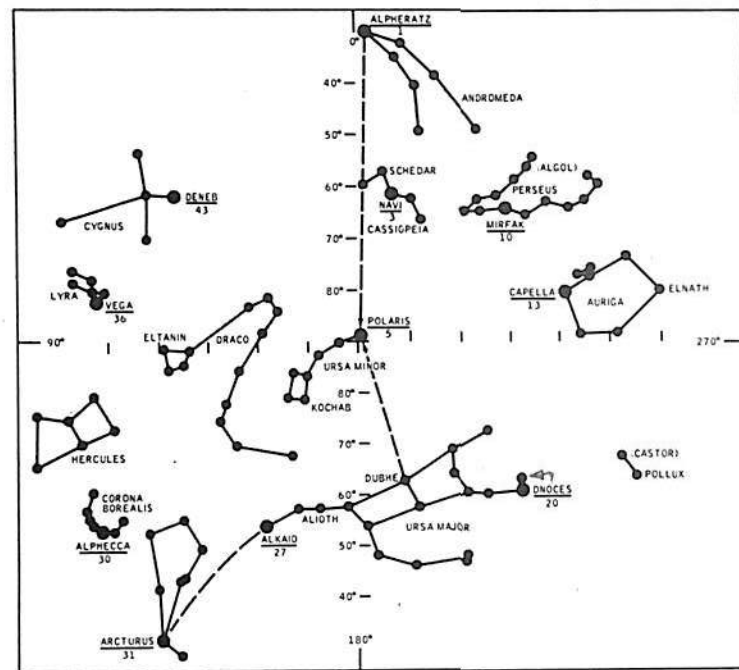
* IVT, ENTRY STATUS EPS, TM, GLYCOL PUMP	1
* C/W, CB ACT, TB, ECS, SUIT FAN/H2O SEP	2
S-BAND & VHF ACT	3
CDR TRANSFER, OPS, DROGUE/PROBE	4
* GLYCOL PUMP, ARS/PGA PRESS	5
REGULATOR CHECK	6
ASC BAT ACT	7
** AOT, S-BD STEERABLE	8
** S-BD BACKUP VOICE, ** S-BD CONFERENCE, ** LTG, COAS CHECK	9
** SEC S-BAND, LANDING GEAR DEPLOY	10
PLSS PREP, PLSS COMM	11
** CSM 1-WAY RELAY, ** LM 1-WAY RELAY	12
** TV/EMU RELAY	13
** LUNAR STAY RELAY, ** LUNAR STAY BACKUP	14
POST PLSS	15
DPS/APS/RCS TEMP/PRESS PGNS ACT, AGS ACT	16







SOUTHERN STARS



NORTHERN STARS

INTRAVEHICULAR TRANSFER TO LM

5 min

- 1 Log ΔT CM Tunnel/Starting ENTRY
STATUS CHECK ()
Actuate CABIN DUMP VALVE & Open Hatch
FLOOD LIGHT - ALL
EXTERIOR LTG - OFF
DES H2O - OPEN
- 2 Docking Tunnel Index Angle $\Delta \phi$ _____
- 3 Transfer ISA To CM
- 4 Transfer LMP Systems Checklist &
Flight Plan Card to LM
Comment On IVT, Restraints &
Tunnel Components

ENTRY STATUS CHECK

10 min

- CB(11)
- Row 1&2: All Open
- Row 3: HEATERS: RNDZ RDR STBY - Close
LDG RDR - Close
LTG: ANUN/DOCK/COMPNT - Close
- Row 4: PGNS: IMU STBY - Close
- Row 5: EPS: BAT FEED TIE (2) - Close
CROSS TIE BAL LOADS - Close
DES ECA - Close
DC BUS VOLT - Close
R/D INST A - Close
DFI PWR - OFF
R/D INST B - Close

- 2 FDAI 1&2 - INRTL
EARTH/LUNAR - PWR OFF
LTG - OFF
MODE - HOLD/FAST
ALT SET - 130

Basic Date Feb. 24, 1969
Changed

LM-3

SYS-2

- 3 FUEL & OXID VENT tb-bp
MASTER ARM - OFF
ASC He SEL - BOTH
LDG GEAR DEPLOY tb-bp
STAGE-SAFE (Guarded)
- 4 S-BAND T/R - OFF (VOL-6)
ICS T/R - OFF (VOL-6)
RELAY - OFF
VOX-ICS (VOX SENS-7)
AUDIO CONT - NORM
VHF A&B - OFF (VOL-6)
MASTER VOL-6
COAS - OFF
- 5 TTCA (CDR) - JETS (DN)
- 6 TIMER CONT - STOP
OVERRIDE ANUN - OFF
OVERRIDE NUM - OFF
OVERRIDE INTEGRAL - OFF
SIDE PANELS - OFF
FLOOD OVHD/FWD - BRIGHT
ANUN/NUM - DIM
INTEGRAL - DIM
- 7 X-POINTER SCALE - HI MULT
RATE/ERR MON - LDG RDR/CMPTR
ATTITUDE MON - PGNS
GUID CONT - PGNS
MODE SEL - LDG RADAR
RNG/ALT MON - ALT/ALT RT
SHFT/TRUN - +50°
RATE SCALE - 25°/SEC
THR CONT - AUTO
MAN THROT - CDR
ENG ARM - OFF
X-TRANSL - 2 JETS
BAL CPL - ON
ASC He REG 1&2 - tb-gray
DESCENT He REG 1 tb-gray
DESCENT He REG 2 tb-bp

Basic Date Feb. 24, 1969

Changed

SYS-3

PRPLNT OTY MON - OFF
 PRPLNT TEMP/PRESS MON - ASC
 HELIUM MON - OFF
 ABORT AND ABORT STAGE - Flush/Guarded

SYS A&B ASC FUEL & ASC OXID (4) - tb-bp
 SYS A&B QUAD 1,2,3,4(8)-tb-gray
 CRSFD - tb-bp
 SYS A&B MAIN SOV - tb-gray
 TEMP/PRESS MON - He
 RCCA - OFF/RESET
 RATE/ERR MON - LDG RDR/CMPTR
 ATTITUDE MON - AGS
 GLYCOL - PUMP 2
 SUIT FAN - 1
 02/H2O OTY MON - ASC 2

ENG GMBL - ENABLE
 DES ENG CMD OVRD - OFF
 LDG ANT - DES
 RADAR TEST - OFF
 TEST MONITOR - ALT XMTR
 SLEW RATE - HI
 RNDZ RDR - AUTO TRACK
 DEAD BAND - MIN
 GYRO TEST - ROLL
 ATTITUDE CONTROL (3) - MODE CONT
 MODE CONTROL - ATT HOLD
 DET - Up And Stop
 TEMP MON - LDG
 RCS SYS A/B-2 QUAD 1,2,3,4 - OFF
 LTG: SIDE PANELS - OFF
 FLOOD - ALL
 OVHD/FWD - BRIGHT
 EXTERIOR LTG - OFF
 LAMP/TONE TEST - OFF
 X-POINTER SCALE - HI MULT

ACA/4 JET (2) - ENABLE
 TTCA/TRANSL (2) - ENABLE
 RDZ ANT RELEASE - STOWED
 AOT - CL, ANGLE - 0000

Basic Date Feb. 24, 1969
 Changed -

9

LM-3

11 TTCA (LMP) - JETS (DN)
AGS STATUS - OFF

12 CB(16)
Row 1: All Open
Row 2: LTG: FLOOD - Close
STAB/CONT: ASA - Close
Row 3: All Open
Row 4: HEATERS: S-BD ANT - Close
EPS: DC BUS VOLT - Close
DES ECA - Close
CROSS TIE BAL LOADS - Close
BAT FEED TIE (2) - Close

13 POWER/TEMP MON - ED/OFF
INVERTER - OFF
BAT 1, 2, 3, 4, - tb-bp
DES BATS - tb-bp
BAT 5&6 NORMAL & BACK UP FEED (4) tb-bp

14 AUDIO CONT - NORM
S-BAND T/R - OFF (VOL-6)
ICS T/R - OFF (VOL-6)
RELAY - OFF
VOX - ICS (VOX SENS-7)
VHF A&B - OFF (VOL-6)
MASTER VOL - 6
VHF A RCVR VOICE BU - OFF

15 S-BAND MODULATE - PM
XMTR/RCVR - OFF
PWR AMPL - OFF
VOICE - OFF
PCM - OFF
RANGE - OFF/RESET
VHF A XMTR & RCVR - OFF (SQUELCH -3)
VHF B XMTR & RCVR - OFF (SQUELCH -3)
TELEMETRY BIOMED - OFF
TELEMETRY - HI
RECORDER - OFF tb-bp
VHF - 1/PLSS TEST
TRACK MODE - OFF
PITCH - +255°
YAW - -30°
S-BAND - 2

Basic Date Feb. 24, 1969
Changed

16

SUIT GAS DIVERTER - PULL/EGRESS
 CABIN REPRESS - CLOSE
 PLSS FILL - CLOSE
 PRESS REG A&B - CLOSE
 DES 02 - CLOSE
 1&2 ASC 02 - CLOSE
 SUIT ISOL (2) - SUIT DISC
 SUIT CIRCUIT RELIEF - AUTO
 CABIN GAS RETURN - AUTO
 CO2 CANISTER SEL - PRIM
 PRIM & SEC CO2 CANISTER - CLOSE
 WATER SEP SEL - PULL/SEP 2
 ASC H2O - CLOSE
 SEC EVAP FLOW - CLOSE
 PRIM EVAP FLOW No. 2 - CLOSE
 DES H2O - OPEN
 WATER TANK SELECT - DES
 SUIT TEMP - COLD
 CABIN TEMP - NORM

CABIN RELIEF AND DUMP FWD-AUTO
 OVHD-OPEN

DFI PRIMARY, SECONDARY (2) - OFF
 UTILITY LIGHTS (Both) - As Required
 FWD Hatch Closed & Locked

EPS, ACTIVATION & CHECKOUT

5 min

- 1 CSM Position LM PWR - RESET Then OFF
 LTG: ANUN/NUM - BRIGHT (1 Caution, 9 Power
 Failure Lts - On)
- 2 CB(11) INST: SIG CONDR 1 - Close
 EPS: XLUNAR BUS TIE - Close
 DES ECA CONT - Close
 SLA/LM ANT TRNFR B&A(2)-Close Then Open
 CB(16) INST: SIG SENSOR - Close
 PCM/TE - Close
 SIG CONDR 2 - Close
 COMM: PRIM S-BD PWR AMPL - Close
 PRIM S-BD XMTR/RCVR - Close

Basic Date Feb. 24, 1969

Changed - 18

LM-3

SYS-6

PMP - Close
 EPS: DISP - Close
 DES ECA CONT - Close
 XLUNAR BUS TIE - Close

3 DFI PRIMARY - ON

4 CB(11) COMM: VHF B XMTR - Close
 VHF B - DATA
 TELEMETRY - LO
 CSM Record LBR Data

5 Verify BAT 1,2,3,4 - LO
 DES BATS - gray
 BATS 5&6 NORMAL & BACKUP (4) - bp
 Check BAT And BUS Voltages (When BUS
 Voltages Less Than 27V Select High
 Voltage Taps)

CB(16)EPS: CROSS TIE BAL
 LOADS - Open
 BAT 1 HI VOLTAGE-OFF/RESET
 tb-bp
 BAT 1 HI VOLTAGE - ON-tb-
 gray
 Repeat For BATS 2,3,4
 CB(16)EPS: CROSS TIE BAL
 LOADS - Close

6 CB(11) AC BUS B&A: BUS TIE INV 2&1 (4)
 -Close
 AC BUS VOLT - Close
 EPS: INV 1 - Close
 CB(16) EPS: INV 2 - Close

7 POWER/TEMP MON - AC BUS
 INV - 1 Then 2
 CB(11) EPS: INV 1 - Open

8 DFI PWR - ON

Basic Date: Feb. 24, 1969
 Changed

ACTIVATE S-BAND FOR TM

1 min

1

S-BAND - PM, PRIM, PRIM, VOICE, PCM, RANGE, OFF, LO
S-BAND ANTENNA - As Desired

*GLYCOL PUMP ACTIVATION

1 min

1

CB(16) ECS: DISP - Close
CB(11) ECS: GLYCOL PUMP AUTO TRNFR-Close
GLYCOL PUMP 1 - Close
GLYCOL PUMP AUTO TRNFR - Open
GLYCOL - PUMP 1
CB(11) ECS: GLYCOL PUMP 2 - Close
GLYCOL PUMP AUTO TRNFR - Close

Basic Date Feb. 24, 1969
Changed

CAUTION/WARNING CHECKOUT

2 min

1 CB(16) INST: CWEA - Close (LGC, CES AC,
CES DC, DES REG, RCS A REG, RCS B REG
Warning Lts, PRE AMPS, HEATER, GLYCOL,
ECS Caution Lts, H2O SEP Comp Lts-On)
LTG: ANUN/DOCK/COMPNT - Close
MASTER ALARM - Close (Master Alarm - On)
HEATERS: DISP - Close

2 RCS TEMP/PRESS MON - Cycle Then He
HTR TEMP MONITOR - Cycle Then LDG
GYRO TEST - POS RT
LAMP/TONE TEST - Check All Positions

CIRCUIT BREAKER ACTIVATION

5 min

1
Row 1: CB(11)
All Closed Except:
AC BUS B&A: AOT LAMP (2) - Open
RNDZ RDR - Open
Row 2: AC BUS A: RNG/RNG RT ALT/ALT RT - Close
All Closed Except:
RCS SYS A: QUAD 4,3,2,1 TCA (4) - Open
Row 3: All Closed Except:
STAB/CONT: ABORT STAGE - Open
(Possible Master Alarm-On With ATCA [PGNS])
AELD - Open
ATT DIR CONT - Open
Row 4: All Closed Except:
ECS: SUIT FAN 1 - Open
PGNS: LDG RDR - Open
RNDZ RDR - Open
(Possible LGC Warning LT & RESTART Lt - On
With LGC/DSKY)
IMU OPR - Open
Row 5: All Closed Except:
EPS: CROSS TIE BUS - Open
ACS ECA CONT - Open
INV 1 - Open
EARTH ORB: SLA/LEM ANT TRNFR (2) - Open

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

SYS-9

2

CB(16)

Row 1: All Closed Except:
RCS: SYS B QUAD 1,2,3,4, TCA - Open

Row 2: All Closed Except:
STAB/CONT: AEA - Open

AELD - Open
ABORT STAGE - Open

Row 3: All Closed Except:
COMM: TV - Open
ECS: GLYCOL PUMP SEC - Open
(Suit Fan Comp Lt - On With Suit Fan
ΔP & Master Alarm May Go On With CO2
SENSOR)

Row 4: All Closed Except:
CAMR: SEQ - Open
EPS: ASC ECA CONT - Open
CROSS TIE BUS - Open

3

MISSION TIMER - SET

4

PRIM EVAP FLOW - OPEN (When Glycol Temp > 60°)

5

RCS SYS A/B-2 QUAD 1,2,3,4 - AUTO

6

CB(16) INST: CWEA - Open & Reclose(Master
Alarm - On RCS A&B REG Warning Lt,
HEATER, ECA, & GLYCOL Caution Lts, & H2O
SEP, SUIT FAN Comp Lt - On)

TB VERIFICATION

1 min

1

FUEL & OXID VENT (2) - bp
LDG GEAR DEPLOY - bp

2

ASCENT He REG 1&2 - gray
DESCENT He REG 1 - gray
DESCENT He REG 2 - bp

3

SYS A&B ASC FUEL & OXID (4) - bp
SYS A&B QUAD ISOL 1,4,2,3(8) - gray
CRSFD - bp
SYS A&B MAIN SOV - gray

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

4

RECORDER TAPE - bp

*ECS ACTIVATION & CHECKOUT

5 min

2

02/H2O QTY MON - ASC 2, ASC 1, DES

SUIT ISOL (Both) - SUIT FLOW

SUIT ISOL (Both) - ACTUATE OVRD (SUIT
Discon)

DES 02 - OPEN

SUIT GAS DIVERTER - PUSH/CABIN

PRESS REG A&B - EGRESS (SUIT Gas Diverter
Automatically Extends And Cabin Fans
Go Off)

SUIT PRESS 4.8 To 5.2 psia

CB(16) ECS: CABIN FAN CONT - Open
(Cabin Fans Go On)SUIT FAN - 2 (Master Alarm, SUIT/FAN
Warning Lt-On Momentarily, ECS
Caution Lt, H2O Sep Comp Lt - On
Then Off In < 2 Min)

PART PRESS CO2 < 7.6 mm Hg

SUIT FAN/H2O SEPARATOR CHECK

2 min

1

CB(16) ECS: SUIT FAN 2 - Open (Master
Alarm, SUIT/FAN Warning Lt, Suit Fan
Comp Lt - On)

2

When ECS Caution & H2O SEP Lt - On, Master
Alarm - On

CB(11) ECS: SUIT FAN 1 - Close

H2O SEP SEL - SEP 1/PUSH

SUIT FAN - 1 (SUIT/FAN Warning Lt, Suit
Fan Comp Lt - Out, ECS Caution Lt,
H2O Sep Comp Lt - Out In < 2 min)

CB(16) ECS: SUIT FAN 2 - Close

3

R/D TLM CAL - ON

DFI PWR - OFF

CB(11):R/D INST A - Open

Basic Date Feb. 24, 1969
Changed

LM-3

S-BAND AND VHF ACTIVATION

(42:57)

Basic Date Feb. 24, 1969

Changed -

LM-3

- 1 Set Comm Configuration
 S-Band - PM, PRIM, PRIM, VOICE, PCM, RANGE, L/R, LO
 VHF - ON, ON, VOICE, ON
 S-BAND & VHF ANTENNA - As Desired
 AUDIO (BOTH):
 S-BAND - T/R (Vol To Just Hear MSFN)
 ICS - T/R
 RELAY - OFF
 AUDIO CONT - NORM
 VOX - ICS
 VHF A (CDR) - T/R
 (LMP) - OFF
 VHF B (CDR) - OFF
 (LMP) - T/R
 Connect Umbilical To Right Side
 (Red/Red - Blue/Blue)
- 2 CSM Configure Simplex B And SET LMP
 Suit Pwr To Off
 Connect To LM Comm Umbilical (Audio, Biomed)
 TAPE - ON
 Conduct COMM Check, Adjust Squelch (ACA &
 Umbilical PTT)
- 3 AUDIO (LMP): VOX - ICS/XMTR
 Conduct Comm Check (Adjust VOX SENS)
 VOX - ICS
- 4 CSM Configure For Simplex A
 AUDIO (LMP): VHF A - T/R
 VHF B - OFF
 Conduct COMM Check, Adjust Squelch
- 5 CSM Configure For LM Data
 VHF B XMTR - DATA
 VHF B RCVR - OFF
 TELEMETRY - LO
- 6 VHF ANTENNA - Select Other Antenna
 Perform COMM Check With CSM

SYS-12

7

CSM Select Other Antenna
Perform COMM Check With CSM

8

VHF ANTENNA - Select Original Antenna
Conduct COMM Check
TAPE - OFF

9

Select Best Antenna Combination

Basic Date Feb. 24, 1969

Changed

CDR TRANSFER AND CREW CONNECT

10 min

- 1 Stow The Following In The ISA
SEQ CAMERA, 5mm & 18mm Lens
CIN168 16mm Mag (R)
Camr Handle
Radiation Survey Meter
Silver Tape
- 2 CSM Set LMP Suit Flow Control To Off
LMP SUIT ISOL - SUIT FLOW (Verify Flow)
Disconnect Umbilical And Pass To CSM
Receive And Stow ISA
LMP Configure Cabin (Strap, Utility
Lights, Cards, Restraints, etc.)
- 3 CDR Transfer To LM With Systems Checklist
Connect To CDR Hoses (Red/Red, Blue/Blue)
CDR SUIT ISOL - SUIT FLOW
Connect To LM Comm Umbilical (Audio, Biomed)
Conduct Comm Check (ACA & Umbilical PTT)
- CDR AUDIO: VOX - ICS/XMTR
Conduct Comm Check (Adjust VOX SENS)
- 5 CDR AUDIO: VOX - ICS
- Interior Photographs Should Be Taken
As Time Available: (CDR Transfer)
For Seq Camera Use CIN168 Film Mag (R),
5mm Lens, f2.8, 1/60, 6 fps
~~For Still Photos Use STD Hasselblad,~~
~~CIN168 Film Mag, f2.8, 1/60, Focus~~
~~As Necessary~~

Basic Date Feb. 24, 1969
Changed

LM-3

OPS CHECK

5 min

- 1 OPS 02 PRESS 5380 To 6380 psi
 02 HOSE - LOCKED
 OPS 02 - ON (02 Press 3.4 To
 4.0 psi)
 HEATER TEST BUTTON - DEPRESS
 At Least 1 Lt - On)
 OPS 02 - OFF
 Verify REG Press Gage Reads Zero In About
 4 min
- 2 Stow OPS (1 On Pallet And 1 On Ascent
 Engine Step)

Install Window Shades And Dark Adapt For
 Daylight AOT Star Visibility Check

DROGUE AND PROBE INSTALLATION

10 min

- 1 Log ΔT Probe/Hatch Closing (_____) 3
 Drogue Lock Lever Engaged And Flush
 Three Capture Latches Engaged And Locked
 (Before And After Preload)
 LM Hatch Exterior Insulation OK, Flaps
 Secured
- 2 Close & Secure Hatch
 CABIN DUMP VALVE - AUTO
 CABIN REPRESS - AUTO
 PRESS REG A&B - CABIN
 CB(16) ECS: CABIN FAN CONT - Close

Basic Date Feb. 24, 1969

Changed

*GLYCOL PUMP CHECK

4 min

- 1 CB(11) R/D INST A - Close
DFI PWR - ON
- CB(11) ECS: GLYCOL PUMP 1 - Open (Master Alarm, And ECS Caution Lt On Momentarily)
CB(11) ECS: GLYCOL PUMP 1 - Close
(Check Glycol Comp Lt - On)
- 3 GLYCOL - INST(SEC) (Note Glycol Press)
CB(16) ECS: GLYCOL PUMP SEC - Close
(10-20 psi Rise) (TBD)
ECS: GLYCOL PUMP SEC - Open
- 4 GLYCOL - PUMP 2 (15-30 psi)
CB(11) ECS: GLYCOL PUMP AUTO TRNFR - Open
GLYCOL - PUMP 1 (Master Alarm & ECS
Caution Lt-On Momentarily, 15-30 psi)
CB(11) ECS: GLYCOL PUMP AUTO TRNFR - Close
- R/D TLM CAL - ON
DFI PWR - OFF
CB(11) R/D INST A - Open

ARS/PGA PRESSURE INTEGRITY CHECK

5 min

- 1 SUIT GAS DIVERTER - PULL/EGRESS
CABIN GAS RETURN - EGRESS
SUIT CIRCUIT RELIEF - CLOSE
PRESS REG A - CLOSE
PRESS REG B - DIRECT 02 (Suit Press To
8.85 psi)
PRESS REG B - CLOSE (Monitor Cuff
Gage Decay < .3 psi In 1 min)
- 2 CO2 CANISTER SEL - SECONDARY (CO2
Comp Lt - On, Monitor Cuff Gage
Decay at < .3 psi In 1 min)

Basic Date Feb. 24, 1969

Changed

LM-3

3 CO2 CANISTER SEL-PRIMARY(CO2 Comp Lt-Off
 SUIT CIRCUIT RELIEF - AUTO
 PRESS REG A&B - CABIN
 CABIN GAS RETURN - AUTO
 SUIT GAS DIVERTER - PUSH/CABIN

4 SUIT TEMP CONTROL
 Position _____
 CABIN TEMP CONTROL
 Position _____

Basic Date Feb. 24, 1969

Changed _____

REGULATOR CHECK

10 min

- 1 Verify CSM Tunnel Hatch Is Closed
- CABIN GAS RETURN - EGRESS
OVHD DUMP VALVE - OPEN
CABIN REPRESS - AUTO
PRESS REG B - EGRESS (Cabin Fans - Off
And Suit Gas Diverter - Egress)
- 3 FWD CABIN DUMP VALVE - OPEN Then AUTO At
4.0psi (Master Alarm, CABIN Warning Light
And Auto Cabin Repress - On At 4.45 psi)
- 4 As Soon As Cabin Repress Starts
PRESS REG A - CLOSE
CABIN Warning Lt - Off, Cabin Repress
Stops)
CABIN REPRESS - CLOSE
FWD CABIN DUMP VALVE - OPEN Then AUTO At
3.5psi (Verify Suit Press 3.6 To 4.0 psi)
- PRESS REG B - CLOSE (Possible Master Alarm,
CABIN Warning Lt-On Momentarily, Cabin
Fans-On)
SUIT CIRCUIT RELIEF - OPEN Then AUTO At
Suit Press of 3.5 psi
PRESS REG B - EGRESS (Suit Press 3.6 To
4.0 psi Master Alarm & CABIN Warning
Lt - On Momentarily, Cabin Fans - Off)
- 6 PRESS REG B - CLOSE (Master Alarm, CABIN
Warning Lt-On Momentarily, Cabin Fans-On)
SUIT CIRCUIT RELIEF - OPEN Then AUTO At
Suit Press of 3.5 psia
PRESS REG A - EGRESS (Suit Press 3.6 To
4.0 psi Master Alarm, CABIN Warning Lt -
On Momentarily, Cabin Fans-Off)
CABIN REPRESS - AUTO
OVHD DUMP - AUTO

Basic Date - Feb. 24, 1969
Changed -

LM-3

7

PRESS REG B - CABIN (Master Alarm, CABIN
Warning Lt, Cabin Fans - On, Suit &
Cabin Press Rises To 4.6 To 5.0 psi)

8

PRESS REG A - CABIN
CABIN GAS RETURN - AUTO
SUIT GAS DIVERter - PUSH/CABIN

Basic Date Feb. 24, 1969

Changed _____

ASCENT BATTERY ACTIVATION & CHECKOUT

- 1 CB(16) EPS: ASC ECA CONT - Close
- 2 POWER/TEMP MON - BAT 5
BAT 5 NORMAL FEED - ON tb-Gray
Verify Bat Current
- 3 POWER/TEMP MON - BAT 6
BAT 6 NORMAL FEED - ON tb-Gray
Verify Bat Current
- 4 BAT 1,2,3,4 HI VOLT - OFF/RESET tb(4)-bp
Verify Bats 1,2,3,4 Current Zero
- 5 BAT 5,6 BACKUP FEED - ON tb(2)-Gray
BAT 5,6 NORMAL FEED - OFF/RESET tb(2)-bp
Verify Bat 5,6 Current
- 6 BAT 1,2,3,4 HI VOLT - ON tb(4)-Gray
Verify Bat 1,2,3,4, Current
BAT 5,6, BACKUP FEED - OFF/RESET tb(2)-bp
Verify Bat 5,6 Current Zero
- 7 CB(16) EPS: ASC ECA CONT - Open
- 8 REPORT ED BAT VOLTAGE TO MSFN
BAT A _____ BAT B _____

DAYLIGHT AOT STAR VISIBILITY

5 min

1

Complete Dark Adaption, CSM MNVR To
NAV STAR Completed
Advise CSM To Disable Thruster B3
CB(11) AC BUS B: AOT LAMP - Close
Unstow AOT & Set AOT Angle - 000.0°
AOT DETENT - F

3

CB(11) AC BUS A: RNDZ RDR - Close
Wait 30 sec
CB(11) PGNS: RNDZ RDR - Close
Unstow RR Antenna & Manually Slew To
AOT Sighting Position
CB(11) PGNS: RNDZ RDR - Open
CB(11) AC BUS A: RNDZ RDR - Open

4

At Daylight Observe Stars
Determine Dimmest Star
RECORD:

GET _____ : _____ : _____
AOT DETENT _____
NAV STAR _____
CSM ICDU ANGLES R _____ P _____ Y _____
DIMMEST STAR _____

5

CB(11) AC BUS B: AOT LAMP - Open
AOT DETENT - CL

6

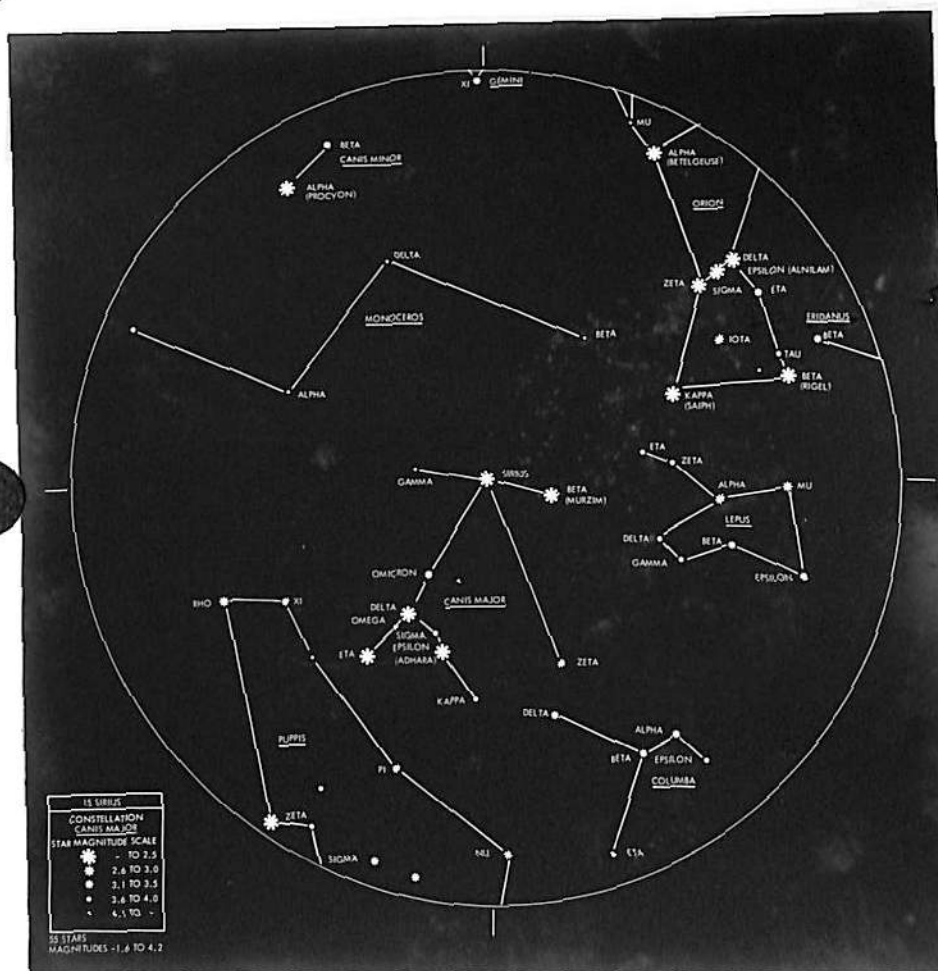
Remove Window Shades

7.

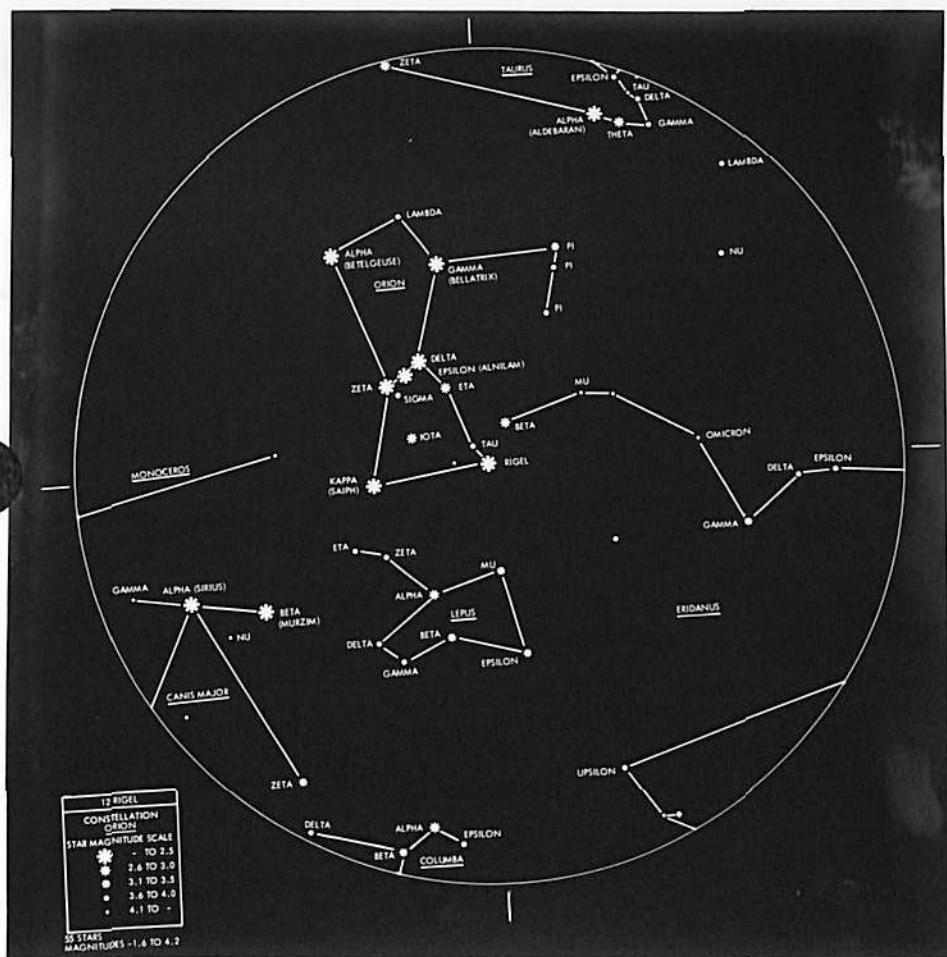
Repower RR And Manually Slew
To 0.0. Remove Antenna Power.

Basic Date Feb. 24, 1969
Changed

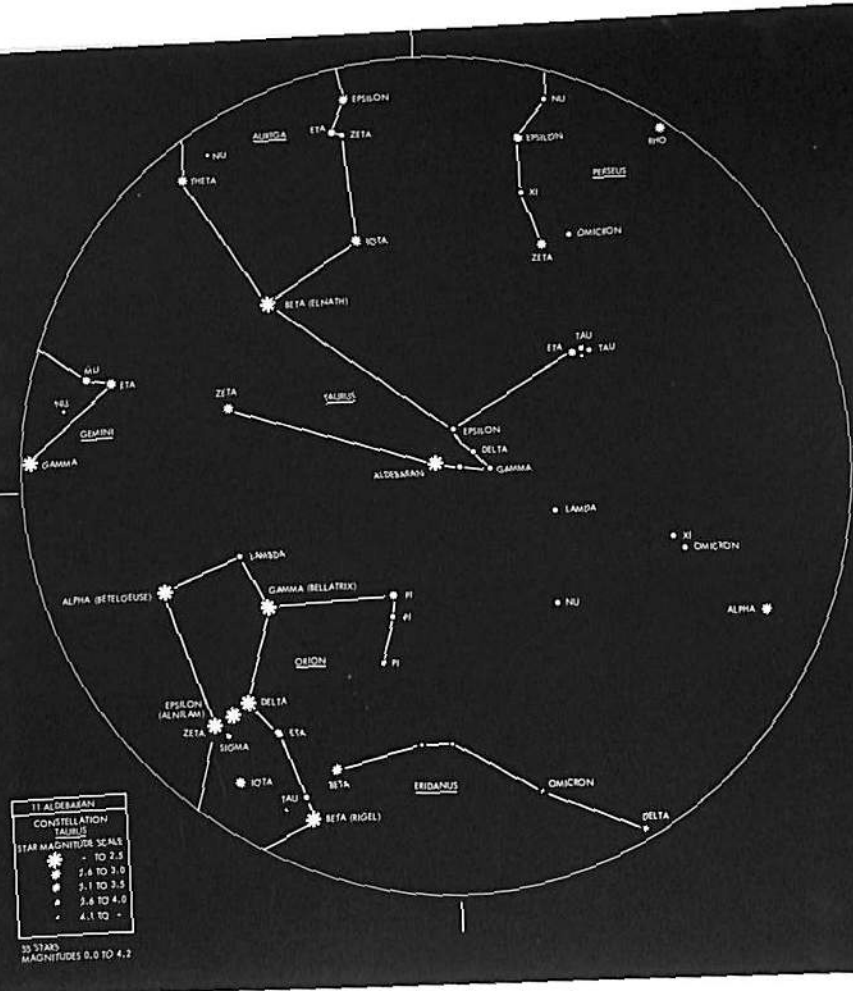
LM-3



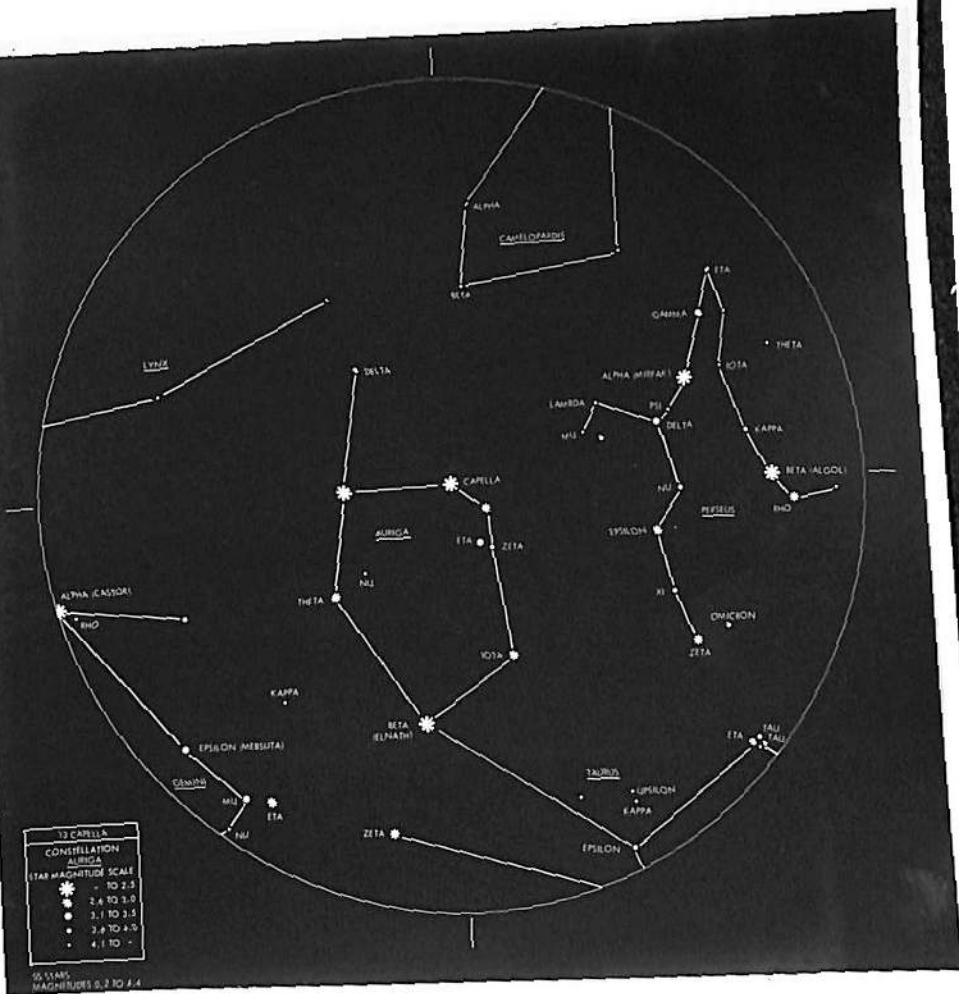
SIRIUS



RIGEL



ALDEBARAN



CAPELLA

**S-BAND STEERABLE ANTENNA CHECK (~44:05 CRO) 8 min

(Docked, Antenna Is Constrained To Elevation Angles Between +100° and +255°, -20° And 75°, Azimuth Angles Between +10° And +75°)

Verify Basic Comm Then
HTR CONT TEMP MONITOR - S-BAND (0°-90°)
S-BAND PWR AMPL - OFF

ANTENNA: S-BAND - SLEW
TRACK MODE - SLEW (To MSFN Update Angles)

Verify MSFN Contact - On Go from MSFN
TAPE - ON

GET _____ PITCH _____ YAW _____
GET _____ PITCH _____ YAW _____
GET _____ PITCH _____ YAW _____

(If No Signal Strength Response, SLEW Prior To T+2 min To New Angles)

Voice Mark Initial Phase Lock, Antenna Angles, And Signal Strength

PITCH _____ YAW _____
SIG. STR. _____ GET _____:_____:_____

When SIG STRENGTH Reaches A Maximum
TRACK MODE - AUTO (Voice MARK)
GET _____:_____:_____

After Detectable Signal Strength Indication
VOICE:

PITCH _____ YAW _____
SIG. STR. _____

Monitor Auto Track For One Minute
VOICE:

PITCH _____ YAW _____
SIG. STR. _____
TAPE - OFF

TRACK MODE - SLEW
~~S-BAND PWR AMPL - PRIM~~

SLEW ANTENNA To PITCH - +225°, YAW - -30°
TRACK MODE - OFF

S-BAND ANT - As Desired
~~S-BAND PWR AMPL - PRIM~~

Basic Date Feb. 24, 1969
Changed

**S-BAND BACKUP VOICE (~44:13 HSK)

4 min

INFLIGHT BU (MODE 4)

Verify Basic Comm Then

PWR AMPL - PRIM

VOICE - VOICE BU

BIOMED - OFF

TELEMETRY - LO

TAPE - ON

Perform Voice And LBR Check With MSFN

TAPE - OFF

Return To Basic Comm

**S-BAND CONFERENCE (MSFN RELAY) (~44:23, MER)

Verify Comm Basic Then

VHF A RCVR - OFF

Increase S-BAND VOLUME

Tape - ON

Perform Voice Check With CSM via MSFN

Tape - OFF

Return To Basic Comm

LIGHTING CHECKOUT

3 min

1 Check All Interior Lighting Normal &
Override
Advise CSM

2 EXTERIOR LTG - TRACK, DOCK, OFF

3 Observe CSM LTG Check

COAS LIGHTING CHECK

1 min

Request CSM Turn On Docking Target

2 COAS-OVHD, Adjust Brightness, Then OFF

3 Advise CSM Test Complete
Adjust COAS For Left Window & Remount
Check Brightness Then OFF

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

LM-3

**SECONDARY S-BAND CHECK (~44:53, ANG) 3 min

- 1 Verify Basic Comm Then On Go From MSFN
S-BAND PWR AMPL - OFF
TAPE - ON
Perform Comm And HBR Check With MSFN (1 min)
- 2 S-BAND PWR AMPL - SEC
S-BAND XMTR/RCVR - SEC
Perform Comm And HBR Check With MSFN (1 min)
- 3 S-BAND XMTR/RCVR - PRIM
S-BAND PWR AMPL - PRIM
Perform Comm Check With MSFN (1 min)
TAPE - OFF

LANDING GEAR DEPLOY

3 min

Unstow, Adjust, And Mount SEQ
Camera Above Right Hand Window
(18mm Lens, CEX368 Mag, 24 fps, f11,
1/250, 20 ft)
CB(11) ED: LOGIC PWR A - Open
R/D INST A - Close
DFI PWR - ON
MASTER ARM - ON
CB(16) CAMR: SEQ - Close
SEQ CAMERA - ON
Observe +Z Gear & Comment On Deployment
And Downlock Shock
LDG GEAR DEPLOY - FIRE: tb-gray
SEQ CAMERA - off
GET: : : (+1.0sec)
CB(16) ED: LOGIC PWR B - Open
CB(11) ED: LOGIC PWR A - Close
LDG GEAR DEPLOY - FIRE
CB(16) ED: LOGIC PWR B - Close
MASTER ARM - OFF (Master Alarm - On)
CB(11) ED: LDG GEAR FLAG - Open
R/D TLM CAL - ON
DFI PWR - OFF
CB(11) R/D INST A - Open
CB(16) CAMR: SEQ CAMR - Open

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

PLSS PREP (~45:05)

20 min

1

Unstow TV, Prepare & Mount (Wide Angle
 Lens, Filter, 10 fps) *Attach Util Lt Brkt,
 Unstow PLSS BATTERIES* *Upper Thumb Fully
 Open, Lower Finger
 Tight*

2

Unstow OPS
 Unstow Antenna Lead From OPS

3

Bleed Down PLSS Hoses By Depressing Check
 Valve Inside Hoses And Restow Hoses
 Remove And Replace Same PLSS Cartridge
 Attach OPS To PLSS
 Connect Antenna Lead To SSC
 Attach PLSS Electrical To SSC
 Install Battery In PLSS
 Connect Battery Cable

4

Unstow RCU & Verify All RCU Controls-OFF
 Connect RCU To PLSS

Verify All PLSS Valves - Up

PLSS COMM CHECK WITH LM (~45:30)

1

Verify Comm Basic

2

VHF B XMTR - VOICE
 VHF B RCVR - ON
 TELEMETRY - HI
 Request CSM Configure VHF A DUPLEX,
 VHF A RCV Only

3

AUDIO (CDR):
 S-BAND - T/R (VOL Full Decrease)
 VOX - ICS
 VHF A - T/R
 VHF B - RCV

Basic Date Feb. 24, 1969
 Changed Feb. 25, 1969

LM-3

- 4 LMP Disconnect LM Comm Cable And Connect
PLSS Comm Umbilical (Audio, Biomed)
- 5 PLSS MODE - POS 1 (Warble Warning Tone On)
TAPE - ON
Perform Comm Check With CDR & Adjust
VOX SENS
Adjust VHF B SQUELCH
- 6 AUDIO (CDR):
VHF A - RCV
VHF B - T/R
- 7 PLSS MODE - POS 4
Perform Comm Check With CDR
Adjust VHF A SQUELCH
TAPE - OFF

Basic Date . 24, 1969
Changed

**CSM 1-WAY RELAY (-45:46 HSK)

5 min

- 1 Request CSM Configure For EVA Comm Mode
- 2 AUDIO (CDR):
S-BAND - RCV (Adjust To Normal Volume)
TAPE - ON
Perform Comm Check With CSM, And MSFN
TAPE - OFF

**LM 1-WAY RELAY (~45:57 MER)

- 1 Verify CB(16) COMM: SE AUDIO - Close
AUDIO (LMP)
S-BAND - T/R
ICS - ~~OFF~~
RELAY - ON
VOX - ICS/XMTR
VHF A - OFF
VHF B - OFF
AUDIO (CDR)
S-BAND - T/R
ICS - T/R
RELAY - OFF
VOX - ICS/XMTR
VHF A - RCV (T/R To Talk)
VHF B - RCV
- 3 PLSS MODE - POS 1
TAPE - ON
Perform Comm Check With CDR & MSFN
(NOTE: CDR Will Have To Relay Success
Of MSFN Reception To LMP)
- 4 LMP Attempt Comm Check With CMP
- 5 CDR Advise On Completion Of Test
Request CSM Return To Comm Basic
LM Return To Comm Basic
TAPE - OFF
PLSS MODE - POS 3
LMP Reconnect To LM Comm Umbilical
(Audio, Biomed)

LM CONFIGURE TO TV/EMC
RELAY (MODE 10)
S-BD MODULATE - FM
AUDIO (LMP) VHF A - T/R
ADJUST VOX & SQUELCH
VERIFY 2-WAY RELAY

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

LM-3

1

Verify Comm Basic, Then
 S-BAND MODULATE - FM *VHF B XMTR-VOICE*
 Prepare And Mount TV Camera (Wide Angle
 Lens (NO Filter), 10 fps) *VHF B RCVR-ON*

AUDIO (CDR):

S-BAND - T/R

ICS - T/R

RELAY - OFF

VOX - ICS/XMTR (Adjust VOX SENS To ~8)

VHF A - RCV (T/R To Talk)

VHF B - RCV

LMP Disconnect From LM Comm Umbilical
 (Audio, Biomed)

Connect To PLSS Comm Umbilical

PLSS MODE - POS 1

CB(16) COMM: SE AUDIO - Close

AUDIO (LMP):

S-BAND - T/R

RELAY - ON *ICS-OFF*

VOX - ICS/XMTR (Adjust VOX SENS To ~8)

MSFN "UP" S-Band Will Key Relay For
 VHF A)

VHF A - T/R (Off At LOS)

VHF B - OFF

VHF A Squelch (Adjust So That CDR Can
 Hear ~~MSFN~~ CSM)VHF B Squelch (Adjusted So CDR Can
 Hear PLSS)

TAPE - ON

Verify 2-Way Relay

LMP Conduct Comm Check With MSFN And CDR
 AUDIO (CDR): Adjust VOX SENS So That CDR &
 PLSS Will Key Relay For "DOWN" S-Band

AUDIO (LMP): Adjust VOX SENS So That MSFN
 "UP" S-Band Will Key Relay For VHF A

On Go From MSFN, CB(16) COMM: TV-Close
 Verify TV Reception With MSFN

At Completion of TV Pass

CB(16) COMM: TV - Open

TAPE - OFF

Basic Date 24, 1969
 Changed Feb. 25, 1969

LM-3

**LUNAR STAY WITH RELAY (MODE 7) (~46:34 BDA)

- 1 MODULATE - PM
PWR AMPL - OFF
RANGE - OFF/RESET
TELEMETRY - LO
TAPE - ON
- 2 LMP Perform Comm, EMU ^{Data} ~~Data~~, And LBR
Data Check With MSFN
TAPE - ON

**LUNAR STAY BACKUP WITH RELAY (MODE 8) (~46:37 ^{VAN} ~~VAN~~)

- 1 PWR AMPL - PRIM
VOICE - VOICE BU
BIOMED - ~~HI~~ ~~LO~~
TELEMETRY - HI
TAPE - ON
- LMP Perform Comm, EMU Data, & HBR Data Check
With MSFN
TAPE - OFF
- 3 Return To Comm Basic
PLSS MODE - POS 3
LMP Disconnect From PLSS Comm Umbilical
And Connect To LM Comm (Audio, Biomed)
Disconnect And Stow TV

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

POST PLSS CHECK (46:45)

15 min

7

Verify All RCU Controls Off
Remove And Stow RCU

Remove Battery Cable
Replace Battery (Verify Locked) And
Stow Used Battery
Stow PLSS Hoses & Electrical Cables
Remove OPS From PLSS And Stow Antenna Lead
Verify All PLSS Valves - Up
Stow PLSS
Stow OPS
Stow TV
Stow Used PLSS BATTERY

DPS/APS/RCS/TEMP/PRESS CHECK

3 min

1 PRPLNT QTY MON - DES 1, DES 2, Then OFF
(Reading Not Valid)

PRPLNT TEMP/PRESS MON

DES 2&1 (50°-75°F, 83-180 psi)

ASC (50°-90°F, 80-200 psi)

3 HELIUM MON

DES AMB PRESS (1430-1750 psi)

DES SUPCRIT PRESS (750-1200 psi)

ASC TEMP 1&2 (30°-140°F)

ASC PRESS 1&2 (2720-3500 psi)

OFF

4 RCS QUANTITY A&B - 100%

RCS TEMP/PRESS MON

He - 2625-3480 psi)

PRPLNT (40°-100°F) (10-130 psi)

FUEL MANF (25-130 psi)

OXID MANF (25-130 psi)

He

5 HTR: RCS TFMP MONITOR - QUAD 1,2,3,4
(40°-190°F),

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

PGNS TURN ON & SELF TEST

4 min

- 1 If STBY Lt-On, PRO
V36E
V21N01E, 3000E, 2176E, E
3011E, 201E, E
1642E, 3777 E
V66E
- 2 CB(11) PGNS: IMU OPR - Close
(NO ATT Lt - On For 90 sec)
- 3 V35E
F 88 88 DSKY LIGHT CHECK
(Master Alarm, LGC, ISS Warning,
And All DSKY Lights - On For 5 sec)
Key RSET
When NO ATT Lt - Off +20 sec, V37E00E
- 4 V25N01E 1365E
F 21 01 E, E, E
- 5 V15N01E 1365E
15 01 R1, R2, R3 All Zero
- 6 V21N27E 10E INITIATE SELF TEST
- 7 KEY REL
15 01 R1 Number Of Errors
R2 Number of Tests Started
R3 Number Of Tests Successful
Test Successful When $R2 \geq 3$ (78sec)
- *PROG Lt - ON *
- *V05N09E 01102 Self *
- * Test Error *
- *N08E Record For MSFN *
- * R1 *
- * R2 *
- * R3 *
- 8 V21N27E 0E TERMINATE TEST

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

AGS ACTIVATION, SELF TEST

2 min

- 1 AGS STATUS - STBY
CB(16) STAB/CONT: AEA - Close
AGS STATUS - OPERATE
(Master Alarm & AGS Warning Lt -
On Then OFF)
- 2 6666 (OPR ERR Lt - On)
- 3 000+88888
- 4 123 - 45679
- 5 412R + 1 SELF TEST SATISFACTORY
+ 3 LOGIC TEST FAILURE
+ 4 MEMORY TEST FAILURE
+ 7 LOGIC AND MEMORY TEST FAILURE
(To Reinitiate Test Set 412 + 0)
- 574R(+) DESCENT STAGE FLAG (+Not Staged)
- 604R(+) LUNAR SURFACE FLAG (+Not On
Lunar Surface)
- 8 612R(+00000) STAGING SEQ COUNTER (+00000
For Att Hold At Abort Stage)

Basic Date Feb. 24, 1969

Changed -

LM-3

**	RATE GYRO, LGC CLOCK, TEPHEM, E-MEMORY DUMP	17
**	DOCKED IMU ALIGNMENT, PIPA BIAS	18
*	RCS PRESS, RCS COLD FIRE	19
**	GIMBAL DRIVE & THROT, RCS HOT FIRE, LGC UPDATE	20
*	LR SELF TEST, RR SELF TEST	21
*	AGS INITIALIZATION, AGS CALIBRATION	22
**	DPS PRESS, IMU REALIGNMENT	23
**	DOCKED DPS BURN	24
	AGS CALIBRATION	25
*	LR SELF TEST	26
	DPS/APS/RCS TEMP/PRESS	27
	CDR TRANSFER	28
	LMP TRANSFER SUBLIMATOR DRYOUT	29
**	INITIAL PWR DOWN, FINAL DEACT	30
	EMERGENCY	31
	UPDATE PADS, PHOTO LOGS, CREW COMMENTS	32

RATE GYRO CHECK

2 min

- 1
 - RATE SCALE - 25°/SEC
 - GYRO TEST - POS RT (Roll Rate +5°/sec)
 - GYRO TEST - NEG RT (Roll Rate -5°/sec)
 - GYRO TEST - PITCH (Repeat Test)
 - GYRO TEST - YAW (Repeat Test)
- 2
 - RATE SCALE - 5°/SEC
 - Repeat Tests

LGC CLOCK INITIALIZATION

5 min

- 1
 - V37E00E
- 2
 - V25N36E Load CSM Time But DO NOT ENTR
 - Seconds
 - ON CSM MARK - ENTR
 - V06N65 (ON CSM MARK - ENTR)
 - COMPUTE CSM/LM Δ TIME
 - Perform Several Times Then
- 4
 - V55E (LOAD Δ TIME)

SET TEPHEM

2 min

- 1
 - CSM V05N01E, 1706E And Read TEPHEM
- 2
 - V25N01E, 1706E LOAD TEPHEM
- 3
 - V05E, 1706E Verify TEPHEM

**ERASABLE MEMORY DUMP

1 min

LM-3  TLM - HI  Verify MSFN Contact
V74E (42 sec, 2 Dumps)

Basic Date Feb. 24, 1969
Changed

LM-3

****LM DOCKED IMU ALIGNMENT**
 (For Same REFSMMAT CSM And LM)

7 min

Verify CSM In MIN DB ATT HOLD
 Until Coarse Align Complete

Calculate LM Gimbal Angles:

OG	IG	MG
300 300 . 26	180 . 00°	
+ ΔΦ		
- CM	+ CM	- CM
LM	LM	LM

V41N20E COARSE ALIGN IMU
 LOAD ICDU ANGLES OG, IG, MG (.01°)
 (NO ATT Lt - On, FDAI Torques)
 *PROG Lt - On *
 *V05N09E 00211 COARSE *
 * ALIGN ERROR, GO To 3*

V40N20E ZERO CDU (NO ATT Lt - Off)
 Notify CSM ATT HOLD No Longer Required

V25N07E
 SET REFSMFLG
 77E, 10000E, 1E
 V01N01E 77E Confirm Bit 13 Is Set

V37E51E
 PRO
 V37E00E

Basic Date Feb 24, 1969
 Changed

LM-3

7

V06N20

06 20

On CSM MARK - ENTR
Copy OG, IG, MG, CSM & LM

OG	IG	MG
. CM	. CM	. CM
. LM	. LM	. LM

Voice Angles To MSFN

8

Copy Ground Calculated Gyro Torquing
Angles

X _____, Y _____, Z _____

9

F 21 93

V42E FINE ALIGN IMU 3 min
LOAD GYRO TORQUING ANGLES X,Y,Z (.001°)

10

16 93

V16N93E
MONITOR TORQUING

Basic Date Feb. 24, 1969
Changed _____

*PIPA BIAS CHECK

4 min

1 DET - Zero
Request CSM Null Rates $<.1^\circ/\text{sec}$ & Go
CMC MODE - FREE

2 V25N21E, E, E, E/Start DET

3 V06E

4 06 21 XYZ PIPA PULSES (+ Pulses)
T+32 - ENTR
T+32 sec(X)R1 _____ (+XXXAB)
(Y)R2 _____
(Z)R3 _____
Notify CSM Test Complete

5 V06N01E, 1452E (R1-Review X BIAS)E (+AB000)
1454E (Review Y BIAS) E
1456E (Review Z BIAS)

F 21 01 V21N01E
LOAD 1452E(Calculated X BIAS) E,E (+AB000)
1454E(Calculated Y BIAS) E,E
1456E(Calculated Z BIAS) E
(Use Same Sign As In Measured BIAS Of
Step 4)

Basic Date - Feb. 24, 1969

Changed -

LM-3

RCS PRESSURIZATION

5 min

Basic Date Feb. 24, 1969
 Changed Feb. 25, 1969

1

RCS QUANTITY A&B - 100%
 Recycle SYS A&B ASC FEED 2(2)-CLOSE
 Recycle SYS A&B ASC FEED 1(2)-OPEN
 ASC FUEL & ASC OXID (4) Remain bp
 SYS A&B QUAD ISOL 1,4,2,3
 tb (8) - gray
 Recycle CRSFD - CLOSE tb-bp
 Recycle SYS A&B MAIN SOV (2)-OPEN tb-gray
 HTR CONT TEMP MON - RCS QUAD 1,2,3,4 >120°F

2

TEMP PRESS MON
 He (2625-3480 psia)
 PRPLNT (40-100°F/10-130 psi)
 FUEL MANF (25-130 psi)
 OXID MANF (25-130 psi)

3

MASTER ARM - ON
 HE PRESS RCS - FIRE
 Recycle SYS A&B ASC FEED 2(2)-CLOSE
 MASTER ARM - OFF (Master Alarm - On, RCS
 A&B REG Warning Lts - Off)

4

RCS TEMP/PRESS MON
 OXID MANF (175-189 psi)
 FUEL MANF (175-189 psi)
 PRPLNT 40°-100°/178-192 psi)
 He (~~2750-3085~~ psi)
 2550-3400

5

Recycle SYS A&B ASC FEED 1(2)-OPEN

6

Recycle SYS A&B QUAD ISOL 1,4,2,3(8)-OPEN
 tb-gray
 ASC FUEL & ASC OXID (4)-tb Remain bp
 Recycle CRSFD - CLOSE tb-bp

RCS COLD FIRE

8 min

- 1 CSM MIN DEAD BAND And ATT HOLD
X-TRANSL - 4 JETS
Verify SYS A&B QUAD ISOL (8) -
OPEN tb-gray
ATTITUDE CONTROL (3) - MODE CONT
MODE CONTROL - ATT HOLD
- 2 V48E LOAD DAP 02012E
- 3 V77E
V15N01E 42E
CDR ACA (TO SOFT STOP)
Roll Right - R3 00051
Roll Left - 77726
Pitch Up - R1 00051
Pitch Down - 77726
YAW Right - R2 77726
YAW Left - 00051
- 4 LMP Repeat Test
- 5 V11N10E 5E
CDR ACA
Roll Right - R1 00245
Roll Left - 00132
Pitch Up - R1 00226
Pitch Down - 00151
CDR TTCA
Up - R1 00252
Down - 00125
- 6 LMP Repeat Test

Basic Date Feb. 24, 1969
Changed

7

E6E

CDR ACA

Yaw Right - R1 00252

Yaw Left - 00125

CDR TTCA

Right - R1 00220

Left - 00140

Fwd - R1 00011

Aft - 00006

8

LMP Repeat Test

9

V76E V11N10E 31E R1 67777

CDR ACA

Roll Right - R1 27757

Roll Left - 27737

Pitch Up - R1 27776

Pitch Down - 27775

Yaw Right - R1 27767

Yaw Left - 27773

LMP Repeat Test

MODE CONTROL - AUTO R1 57777

Basic Date Feb. 24, 1969

Changed

11

LM-3

**GIMBAL DRIVE AND THROTTLE TEST

4 min

Basic Date Feb. 24, 1969
Changed

LM-3

- 1 THR CONT - MAN
MAN THROT - SE
TTCA (Both) - THROTTLE (UP) MIN POSITION
- 2 Verify MSFN Contact
V48E
F 01 46 DAP CONFIGURATION
R1 02012
PRO
- 3 F 06 47 LM WT, CSM WT (1b)
R1 _____ (+32428)
R2 _____ (+28788)
PRO
- 4 F 06 48 ENGINE GIMBAL TRIM PITCH, ROLL (.01°)
R1 + _____ (~~00610~~) (+00582)
R2 + _____ (~~00675~~) (+00681)
Verify MODE CONTROL - AUTO
ENGINE GMBL - ENABLE
ENGINE ARM - DES (Master Alarm, DES REG - On)
PRO (Master Alarm And GDA/RCCA Lt-On When
Gimbals Reach Limits)
- 5 TTCA (LMP) - MIN, Then SOFT STOP (51%),
Then MAX (Off Scale High), Then MIN
MAN THROT - CDR
TTCA (CDR) - MIN, Then SOFT STOP (51%),
Then MAX (Off Scale High), Then MIN
- 6 THR CONT - AUTO
- 7 F 50 48 TRIM COMPLETE
ENG ARM - OFF (GDA/RCCA Lt-Off)
TTCA (Both) - JETS (DN)
PRO
MSFN Verifies Final GDA Position If
Gimbal Angles Differ From Desired
Values By More Than 0.1° Repeat V48

**RCS HOT FIRE

7 min

- 1
 - Verify HBR With MSFN
 - Verify RCS QUAD TEMP 1,2,3,4, (120°-190°F
 - For At Least 25 min
 - CB(16) INST: CWEA - Open & Reclose
 - (Master Alarm - On)
 - GUID CONT - AGS
 - X-TRANSL - 4 JETS
 - DEAD BAND - MAX
 - ATTITUDE CONTROL (3) - MODE CONT
 - MODE CONTROL - AFT HOLD
- 2
 - CDR ACA (Deflect Slowly To Soft Stop,
 - Pause At Null)
 - Roll Right, Left, Pitch Up, Down, Yaw
 - Right, Left (Possible Master Alarm, RCS
 - TCA Caution Lt - On tb-Red)
 - Verify MSFN Read RCS Data
 - ATTITUDE CONTROL (3) - PULSE
 - CB(11) STAB/CONT: ATT DIR CONT - Close
 - CDR ACA (Deflect Hardover, Pause At Null)
 - Roll Right, Left, Pitch Up, Down, Yaw
 - Right, Left (Possible Master Alarm & RCS
 - TCA Caution Lt-On tb-Red
- 4
 - CB(11) RCS: SYS A QUAD 4,3,2,1 - Close
 - CB(16) RCS: SYS B QUAD 1,2,3,4, - Close
 - CB(16) INST: CWEA - Open & Reclose
 - CDR TTCA
 - Up, Down, Right, Left, Fwd, Aft
- 5
 - V76E
 - GUID CONT - PGNS
 - CDR TTCA (Pause At Null)
 - Up, Down, Right, Left, Fwd, Aft
 - Standby For MSFN GO
- 6
 - X-TRANS - 2 JETS
 - DEADBAND - MIN
- 7
 - Set DAP - 11001

3 min

MSFN Uplink REFSMMAT & State Vector &
NAV CHECK PAD (Uplink Acty Lt - On
Then Off When Uplink Completed)

3 Copy Nav Pad

												T
												$\phi(+N)$
												$\lambda(+E)$
												NAV
												CHECK
												H

*LANDING RADAR SELF TEST

4 min

- 1 Log ALT/ALT RT & X-POINTERS During Test
 Verify MSFN Contact
 X-POINTERS(Both) - HI MULT
 RATE/ERR MON - LDG RDR/CMPTR
 TEMP MON - LDG ($>49^{\circ}$) ~~(180)~~
 RNG/ALT MON - ALT/ALT RT
 LDG ANT - DES
 MODE SEL - LDG RDR
 CB(11) PGNS: LDG RDR - Close
 (X-POINTERS Will Oscillate Then
 Up And Right Off Scale)
~~CB(11) E/D INST A - Close~~
~~DFI PWR - ON~~
- 2 RADAR TEST - LDG(Alt And Alt Rt Tapes
 Drive)
 TEST MONITOR-ALT XMTR 2.1 To 5.0v(3.6v)()
 -VEL XMTR 2.1 To 5.0v(3.8v)()
 ALT/ALT RT MON - 8094 To 8457 ft/-433 To
 -465 fps, (8280/-450) ()
- LDG ANT - HOVER (10 sec)
 ALT/ALT RT - +7818 To 8169 ft/-441 To
 -457 fps (8000/-450) ()
- LDG ANT - DES (Wait 10 sec)
- 5 V62E INITIATE RDR SELF TEST
 R1 00004 SPECIFY RDR
 R2 00002 LDG RDR
 PRO
- 6 F 04 06
- SLANT RANGE, ANT POSITION (ft)
 R1 +08156 To +08418 (+08286) ()
 R2 +00001
 PRO
- LM-3 F 16 67
- LDG RDR VEL X,Y,Z (fps)
 R1 -00230 to -00264 (-246) ()
 R2 +00924 to +00954 (+930) ()
 R3 +00643 to +00689 (+666) ()

Basic Date — Feb. 24, 1969
 Changed — Feb. 25, 1969

SYS-49

8 V34E

9 LDG ANT - AUTO

10 V61E COMMAND ANT TO POS 2 (27 sec)
ALT/ALT RT MON - +7818 To +8169/-441
To -457 fps (8000/-450) ()

11 F 04 06 V62E INITIATE RDR SELF TEST
R1 00004 SPECIFY RDR
R2 00002 LDG RDR
PRO

12 F 16 66 SLANT RANGE, ANT POSITION (ft)
R1 +08156 to +08418 (+08292) ()
R2 +00002

13 F 16 66 LDG ANT - DES (10 sec)
R2 +00001 (PRO Lt - On, V05N09E, 00522)

14 LDG ANT - AUTO
V34E
RDR TEST - OFF

15 V78E START LR SPUR TEST

16 V79E STOP LDG RDR SPUR TEST

17 CB(11) PGNS: LDG RDR - OPEN (Master
Alarm - On)

18 BID TLM CAL - ON
DPI PWR - OFF
CIB(11) BID INST A - Open

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

*RNDZ RDR SELF TEST

10 min

1 Verify CSM RCS Thruster B3 And RR Xponder
- OFF

RDZ ANT - Released

X-POINTERS (Both) - HI MULT

RATE/ERR MON (Both) - RNDZ RADAR

ATTITUDE MON (Both) - PGNS

MODE SEL - LDG RADAR

RNG/ALT MON - RNG/RNG RT

SHAFT/TRUN - +50°

RDZ RDR - SLEW

TEMP MONITOR - RNDZ (+10° To °150°)

~~CB(11) R/D INST A - Close~~

~~DFT PWR ON~~

2 CB(11) AC BUS A: RNDZ RDR-Close (Wait
30 sec)

PGNS: RNDZ RDR - Close(NO TRACK Lt-On)

SLEW LEFT TO 0°, 0°

SLEW RATE - LO

SHFT/TRUN - +5°

Slew Antenna Up, Down, Left, Right To Verify
Slew

RDZ RDR - AUTO TRACK (Master Alarm And
RNDZ RDR Caut Lt - On)

RADAR TEST - RNDZ (Rng Rt Tape Drives,
X-Pointers and FDAI Needles Vary Between
Limits. After 12 sec, Rng Tape Drives,
NO TRACK And RNDZ RDR Caut Lt - Out)

5 TEST MONITOR-AGC .7 to 3.0V(1.5)()

-XMTR PWR 2.1 to 4.8v(2.8)()

-SHAFT ERR 1.5 to 3.5v

(~~1.5~~ to ~~3.5~~)()

-TRUN ERR 1.5 to 3.5v

(~~1.5~~ to ~~3.5~~)()

-AGC 2.2 2.4 ()

RDZ RDR - SLEW

Slew Antenna To 0°, 0°

RNDZ RDR - LGC (NO TRACK Lt - On)

Basic Date Feb. 24, 1969
Changed LM-3

LM-3

- 6 F 04 06 V62E START RNDZ RDR SELF TEST
R1 00004 SPECIFY RADAR
R2 00001 RNDZ RADAR
PRO
(NO TRACK Lt - Out After 12 sec)
- 7 F 16 72 RR TRUNNION AND SHAFT (.01°)
R1 Varying @ 1/2 cps
R2 Varying @ 1/2 cps
PRO
- 8 F 16 78 RANGE, RANGE RATE (.01nm, fns)
R1 +18900 To +19800 (+19571)
R2 -00459 To -00541 (-00495)
RNG/RNG RT MON +189 To +198 nm/-459 To
-541 fps (196/-493) ()
- 9 V34E
- 10 RADAR TEST - OFF (NO TRACK Lt - On, X-PNTR-
Center)
- 11 V40N72E RR CDU ZERO (10 sec)
- 12 V41N72E
N73 R1+04000
R2+04000
N06 R2 00002
V16N72E (Verify FDAI Needles Up & Right)
V44E (TERM DESIG)
- 13 V41N72E
N73 R1-00400
R2-00400
N06 R2 00002
V16N72E (Verify FDAI Needles)
V44E (TERM DESIG)

Basic Date Feb. 24, 1969
Changed _____

~~14~~~~V14N72E~~~~N73 R1=00000~~~~R2=00000~~~~N06 R2=00002~~~~V16N72E (Verify FDAI Needles)~~~~V44E (TERM DESIG)~~

15 14

V41N72E

N73 R1+00000

R2 ~~28300~~ +00000

N06 R2 00002

V16N72E

CB(11) PGNS: RNDZ RDR - Open

AC BUS A: RNDZ RDR - Open

V44E (TERM DESIG)

~~16 15~~~~R/D TLM CAL ON~~~~DEF PWR - OFF~~~~CB(11) R/D INST A Open~~Basic Date Feb. 24, 1969
Changed

4

V40N20E ZERO CDU

5

400 + 6E CALIBRATE GYRO & ACCEL

Read And Record After 32 sec

540R X ACCEL BIAS COEFF _____ (Octal)

541R Y ACCEL BIAS COEFF _____ (Octal)

542R Z ACCEL BIAS COEFF _____ (Octal)

6

If It Appears That The Gimbal Angles Will
Pass Thru 0°, 45°, 90°, 135°, 180°, 225°, 270°,
Or 325°, Exit Calibration By 400+0E

7

400R+0 When GYRO & ACCEL CALIBRATE Complete

8

Read And Record After 5 min 2 sec

544R X GYRO DRIFT COEFF _____ (.01°/hr)

545R Y GYRO DRIFT COEFF _____ (.01°/hr)

546R Z GYRO DRIFT COEFF _____ (.01°/hr)

DPS PRESSURIZATION

4 min

- 1 PRPLNT TEMP/PRESS MON
ASC-(50°-90°F/80-200 psi)
DES 1&2 (50°-75°F/83-180 psi)
- 2 HELIUM MON:
AMB PRESS (1430-1750 psi)
SUPCRIT PRESS (715-1200 psi)
TEMP 1&2 (30°-140°F)
PRESS 1&2 (2720-3500 psi)
AMB PRESS
- 3 DES He REG 1 tb-gray
DES He REG 2 - tb-bp
- 4 MASTER ARM - ON
DES PRPLNT ISOL VLV - FIRE
He PRESS DES START - FIRE
MASTER ARM - OFF (Master Alarm - On)
- PRPLNT TEMP/PRESS MON:
DES 2&1 (50°-75°F/242-253 psi)
- HELIUM MON:
DES AMB PRESS (500-1700psi)
DES SUPCRIT PRESS (715-1200 psi)

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

LM-3

23

**LM DOCKED IMU REALIGNMENT

5 min

- 1 Verify CSM Has Completed Final P52
V06N20E
06 20 On CSM MARK - ENTR
Copy OG, IG, MG, CSM & LM

OG	IG	MG
. CM	. CM	. CM
. LM	. LM	. LM

23

Voice Angles To MSFN

- 2 Copy Ground Calculated Gyro Torquing
Angles

X____, Y____, Z____

- 3 V42E FINE ALIGN IMU
F 21 93 LOAD GYRO TORQUING ANGLES (.001°)

- 4 V16N93E
F 16 93 MONITOR TORQUING

Basic Date Feb. 24, 1969
Changed

LM-

**DOCKED DPS BURN

20 min

1

~~DET~~ ~~ON~~

CEX 368 MAG(S)

Adjust SEQ CAMR (12 fps, 1/250, f8, 18ft) And
Remount In R.H. Window (~~18ft~~)

V37E 00E

V76E

V63E MODE II NEEDLES

Verify DAP - 11001

GUID CONT - PGNS

MODE SEL - LDG RADAR

RNG/ALT MON - ALT/ALT RT

RATE/ERR MON - LDG RDR/CMPTR

X-POINTERS (Both) - HI MULT

DEAD BAND - MIN

ATTITUDE CONTROL (3) - MODE CONT

MODE CONTROL - ATT HOLD

2

V37E 30E

N33 _____ TIG

N82 _____ ΔVX

_____ ΔVY

_____ ΔVZ

N42 _____ HA

_____ HP

_____ ΔVR

N45 M, TFI, MGA

SET DET

3

V37E 00E

Update AGS

Align AGS

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

LM-3

24

4

V37E 40E
 V65E DISABLE UV JETS
 V06 N86 _____ ΔV_X
 _____ ΔV_Y
 _____ ΔV_Z
 410 + 5
 450 _____ ΔV_X
 451 _____ ΔV_Y
 452 _____ ΔV_Z
 267 R _____
 411 + 0
 407 + 0
 400 + 1
 407R
 KEY REL

F 50 18 ENTR

5

Request CSM Maneuver To Burn Attitude
 (300, 180, 0)

Monitor Maneuver On LM FDAI To 0,0,0

Request CSM Go To CMC Free

MODE CONTROL - AUTO

F 50 19

ENTR

CB(11) AC BUS B: AOT LAMP - Close

Verify AOT Detent & Set Index - 000.0°

Verify Burn Attitude Star (GO-NO GO

Limit +5°)

CB(11) AC BUS B: AOT LAMP - Open

AOT Detent - CL

PRO

BURN MONITOR TECHNIQUE

The Established Attitude Limit Is +10° And
 Rate Limit Is +5°/sec. Because It Is Pro-
 bable That There Could Be Some Greater Ex-
 *cursion During Thrust Transight Periods, *
 *The Absolute Attitude Limit Is +45°, This *
 *Should Be Coincident With Program Alarm *
 *1407 (VG INCREASING). The GDA Light Is To *
 *Be Essentially Ignored During The Burn *

(During Burn Comment On X-POINTER & ALT/ALT
 RT Meter Characteristics)

Basic Date 0 24, 1969
 Changed _____

LM-

6

CB(11) PGNS: LDG RDR - Close
 V78E Start SPURIOUS TEST
 TEMP MONITOR - LDG RADAR ()

-9:00

~~PRPLNT QTY MON - DES 1,2~~
 CB(11) R/D INST A - Close
 DFI PWR - ON
 TAPE - ON

-6:00

~~-6:00~~

~~PRPLNT QTY MON - DES 1,2~~
 CB(11) EPS: INV 1 - Close
 INV - 1 (Possible Master Alarm - On)
 CB(16) EPS: CROSS TIE BAL LOADS - Open

-4:00

TTCA (CDR) - THROTTLE (UP)
 THR CONT - MAN
 MAN THROT - CDR
 Verify MODE CONTROL - AUTO
 ENG ARM - DES
 X-TRANSL - 2 JETS
 BAL CPL - ON
 ENG GMBL - ENABLE
 PRPLNT TEMP/PRESS MON - DES 1
 HELIUM MON - SUPCRIT PRESS
 CAMERA SEQ - Close
 MASTER ARM - ON
 Verify MAN THROTTLE Set To Min Then
 THR CONT - AUTO

-2:00

Verify AGS Attitude Errors
 500R VGX

-:15

SEQ CAMERA - ON

-:10

Begin MANUAL ULLAGE

-:05

F 99 40
 RELEASE ULLAGE
 PRO

-0:00

IGNITION (Do Not Reset DET)

59:57

Verify Squibs Blown
 (SUPCRIT PRESS Increasing)
 MASTER ARM - OFF (Master Alarm - On)

CALL-	
DET	OUT
2 @	2
7 @	5
12 @	5
17 @	5
22 @	5
24 @	2
29 @	5
31 @	2

Basic Date — Feb. 24, 1969
 Changed — Feb. 25, 1969

LM-3

SYS-60

59:55 THROTTLE TO 40%
(Not Less Than Ignition +5 sec)

59:34 Monitor Auto Throttle To FTP

59:00 SEQ CAMERA - OFF
VG+200 fps SEQ CAMERA - ON
Monitor DES QTY MON And SUPCRIT
PRESS (> 350 psi)

VG+124 fps THR CONT - MAN
EXECUTE THROTTLE PROFILE

TFC = 10 DES He REG 1 - Close
TFC = 03 ENG STOP PB - DEPRESS
MODE CONTROL - ATT HOLD
SEQ CAMERA - OFF

Request CSM Resume ATT HOLD (If Excessive
Rates Exist They Should First Be Damped
By LM Y and Z Translational Thrusters)
ENG ARM - OFF

PRO
N85 _____ VX
_____ VY
_____ VZ

PRO
V79E TERM SPURIOUS TEST
00E

F 37

TTCA (CDR) - JETS (DN)
Verify ENG ARM - OFF
ENGINE STOP PB - RESET
THR CONT - AUTO
LDG RDR TEMP (_____)
PRPLNT QTY MON - OFF
INV -2 (Possible Master Alarm - On)
CB(11) EPS: INV 1 - Open
CB(16) EPS: CROSS TIE BAL LOADS - Clos
CAMR: SEQ - Open

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

10

500R _____ VX
 501R _____ VY
 502R _____ VZ

V82E

N06 R2 00001

PRO

N44 _____ HA
 _____ HP

12 CUT

OFF +2min

Landing Radar Temp _____

GET _____:_____:

Comment On:

Visual Affect Of Plume

Noise & Vibration

Exhaust On LDG GEAR

Vehicle Dynamics

13

R/D TLM CAL - ON

DFI PWR - OFF

TAPE - OFF

CB(11) PGNS: LDG ZDR - Open (Master Alarm-
 On)

Basic Date Feb. 24, 1969

Changed -

LM-3

AGS CALIBRATION

9 min

1 Read And Record
 540R X ACCEL BIAS COEFF _____ (Octal)
 541R Y ACCEL BIAS COEFF _____ (Octal)
 542R Z ACCEL BIAS COEFF _____ (Octal)

 544R X GYRO DRIFT COEFF _____ (.01°/hr)
 545R Y GYRO DRIFT COEFF _____ (.01°/hr)
 546R Z GYRO DRIFT COEFF _____ (.01°/hr)

2 Verify AGS in Standby/Operate For 25 min
 PGNS-ON, LM Thrusters Disabled,
 Rates < .1°/sec, RPY ICDUS Torqued
 Beyond 11.25° And Will Not Pass Thru
 0°, 45°, 90°, etc. (CSM OG=82.5°, IG=22.5°,
 MG=22.5° Will Give The Desired Starting
 Attitude)

3 CSM Establish AGS Calibration Attitude, Mini-
 mize Rates, Then Go CMC MODE-FREE
 Monitor ICDU Angles (All Angles Should
 be Approx. 22°, 67°, 112°, 157°, 202°, 247°,
 292°, Or 337°)
 LM ICDUS: ~~OG~~ 112.5°
 ~~IG~~ 202.5°
 ~~MG~~ 022.5°

FDAI ANGLES: R 132.7°
 P 339.8°
 Y 301.4°

4 V40N20E ZERO CDU

5 400 + 6E CALIBRATE GYRO & ACCEL
 ATTITUDE MON-Cycle To Verify AGS Alignment
 Read And Record After 32sec
 540R X ACCEL BIAS COEFF _____ (Octal)
 541R Y ACCEL BIAS COEFF _____ (Octal)
 542R Z ACCEL BIAS COEFF _____ (Octal)

Basic Date Feb. 24, 1969Changed

LM-3

25

6 If It Appears That The Gimbal Angles Will
Pass Thru 0° , 45° , 90° , 135° , 180° , 225° , 270° ,
 325° , Exit Calibration By 400+0E

7 400R+0 When GYRO & ACCEL CALIBRATE COMPLETE

8 Read And Record After 5 min 2 sec

544R X GYRO DRIFT COEFF _____ ($.01^{\circ}/\text{hr}$)

545R Y GYRO DRIFT COEFF _____ ($.01^{\circ}/\text{hr}$)

546R Z GYRO DRIFT COEFF _____ ($.01^{\circ}/\text{hr}$)

25

Basic Dat. Feb. 24, 1969
Changed _____

3

*LANDING RADAR SELF TEST

4 min

- 1 Verify MSFN Contact
 X-POINTERS(Both) - HI MULT
 RATE/ERR MON - LDG RDR/CMPTR
 TEMP MON - LDG (>49°F)
 RING/ALT MON - ALT/ALT RT
 LDG ANT - DES
 MODE SEL - LDG RDR
 CB(11) PGNS: LDG RDR - Close
 (X-POINTERS Will Oscillate Then
 Off Scale Up And Right)
- 2 RADAR TEST - LDG(Alt And Alt Rt Tapes
 Drive)
 TEST MONITOR-ALT XMTR 2.1 To 5.0v(3.6v)()
 -VEL XMTR 2.1 To 5.0v(3.8v)()
 ALT/ALT RT MON - +8094 To +8457 ft/433 To
 -465 fps (8280/-450) ()
- LDG ANT - HOVER (10 sec)
 ALT/ALT RT - +7818 To +8169 ft/-441 To
 -457 (8000/-450) ()
- LDG ANT - DES (Wait 10 sec)
- V62E INITIATE RDR SELF TEST
 R1 00004 SPECIFY RDR
 R2 00002 LDG RDR
 PR0
- 6 F 16 66 SLANT RANGE, ANT POSITION (ft).
 R1 +08156 To +08418 (08286) ()
 R2 +00001
 PR0
- F 16 67 LDG RDR VEL X,Y,Z (fps)
 R1 -00230 to -00264 (-00246) ()
 R2 +00924 to +00954 (+00930) ()
 R3 +00643 to +00689 (+00666) ()

26

Basic Date Feb. 24, 1969

Changed

4

5

LM-3

SYS-65

- 8 V34E
- 9 LDG ANT - AUTO
- 10 V61E COMMAND ANT TO POS 2 (27 sec)
ALT/ALT RT MON - +7818 To +8169/-441 To
-457 fps (8000/-450) ()
- 11 F 04 06 V62E INITIATE RDR SELF TEST
R1 00004 SPECIFY RDR
R2 00002 LDG RDR
PRO
- 12 F 16 66 SLANT RANGE, ANT POSITION (ft)
R1 +08156 To +08418 (08292) ()
R2 +00002
- 13 F 16 66 LDG ANT - DES (10 sec)
R2+00001 (PROG Lt-On, V05N09E, 00522)
- 14 LDG ANT - AUTO
V34E
- 15 RDR TEST - OFF
CB(11) PGNS: LDG RDR-OPEN(Master Alarm-On)

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

Li

DPS/APS/RCS TEMP/PRESS CHECK

2 min

- 1 PRPLNT TEMP/PRESS MON -
 DES 2&1 (50°-75°F/238-253 psi)
 ASC (50°-90°F/80-200)

HELIUM MON:

DES AMB PRESS (238-253)
 DES SUPCRIT PRESS (900-1100)
 ASC TEMP 1&2 (30°-140°F)
 ASC PRESS 1&2 (2720-3500 psi)
 DES SUPCRIT PRESS

- 3 RCS QUANTITY A&B (95%)
 RCS TEMP/PRESS MON:
 He (2475-3325)
 PRPLNT (40°-100°F/178-192 psi)
 FUEL MANF 175-189 psi)
 OXID MANF (175-189 psi)
 He

HTR: RCS TEMP MONITOR - QUAD 1,2,3,4
 (120°-190°F)

SUBLIMATER DRYOUT 53 min To Final Pwr Down

- 1 V37E06E
 F 50 25 R1 00062 POWER DOWN LGC
 CB(11) PGNS IMU OPR - Open
 PRO (STBY Lt-On)
- 2 CB(16) STAB/CONT: AEA - Open
 AGS STATUS - OFF
 DFI PWR - ON
- 3 PRIM EVAP FLOW - Close
 (Dryout Complete In 53 Min) ~~Glycol Temp~~
~~Should Not Go Above 95°(TM)~~
 DET - RESET/START

Basic Date Feb. 24, 1969
 Changed Feb. 25, 1969

LM-3

27

CDR TRANSFER TO CSM

10 min

- 1 Stow The Following In The ISA;
 Exposed Film (2/16mm Mags, (S)&(R)
 Leave 70mm Mag (F) In LM
 SWA Hasselblad With Camr Handle
 Seq Camera, 5mm & 18mm Lens
 EVA Checklist (2) & Flight Plan Card
 Verify Tunnel Pressurized
 PRESS REG A&B - EGRESS
 OVHD CABIN DUMP VALVE - OPEN
 Stow OPS On Floor Note Press (5380-6380 psi)
 Open Hatch
 OVHD CABIN DUMP VALVE - AUTO
 Log ΔT Hatch Opening/Tunnel Clear (____)
 Assist CMP In Clearing Drogue/Probe From
 Tunnel

- 2 CDR SUIT ISOL - SUIT DISC
 Disconnect LM Hoses And Stow (Audio, Biomed)
 CDR Transfer To CSM With ISA & CDR
 SYSTEMS Checklist
 Log ΔT CDR At Crew Station/Ingress CSM (____)

Basic Date _____
 Changed _____
 Feb. 25, 1969

LMP TRANSFER TO CSM UMIBILICAL

2 min

- 1 Connect CSM Umbilical
 LMP SUIT ISOL - SUIT DISC
 Request CSM Set LMP Suit Flow Control To
 Suit Full Flow And Audio Suit Power - Off
- 2 Connect To CSM Comm (AUDIO, BIOMED), CDR
 Notify CMP Then Conduct Comm Check
- 3 Disconnect LM Hoses And Stow
 DES 02 - CLOSE
 PRESS REG A&B - CLOSE

Basic Date Feb. 24, 1969
Changed Feb. 25, 1969

LV-3

INITIAL POWER DOWN

- 1 FDAI 1 & 2 - INRTL
EARTH/LUNAR - PWR OFF
LIGHTING - OFF
MODE - HOLD/FAST
ALT SET - 130
- 2 MASTER ARM - OFF
ASC He SEL - BOTH
STAGE - SAFE/GUARDED
- 3 AUDIO CONT - All Switches-OFF, COAS - OFF
- 4 TTCA (CDR) - JETS (DN)
- 5 OVERRIDE ANUN - OFF
OVERRIDE NUM - OFF
OVERRIDE INTEGRAL - OFF
SIDE PANELS - OFF
INTEGRAL - DIM

X-POINTERS SCALE - HI MULT
RATE/ERR MON - LDG RDR/CMPTR
ATTITUDE MON - PGNS
GUID CONT - PGNS
MODE SEL - LDG RADAR
RNG/ALT MON - ALT/ALT RT
SHFT/TRUN - +50°
RATE SCALE - 25°/sec
THR CONT - AUTO
MAN THROT - CDR
ENG ARM - OFF
X-TRANSL - 2 JETS
BAL CPL - ON
ASC He REG 1 & 2 - OPEN tb-gray
DESCENT He REG 1 tb-bp
DESCENT He REG 2 tb-bp
PRPLNT QTY MON - OFF
PRPLNT TEMP/PRESS MON - ASC
HELIUM MON - OFF
ABORT And ABORT STAGE - Flush/Guarded

Basic Date Feb. 24, 1969
Changed

SYS-70

7
 SYS A&B ASC FUEL & ASC OXID (4) tb-bp
 SYS A&B QUAD 1,2,3,4 (8) - OPEN tb-gray
 CRSFD tb-bp
 SYS A&B MAIN SOV tb-gray
 RCS TEMP/PRESS MON - He
 RCCA - OFF/RESET
 RATE/ERR MON - LDG RDR/CMPTR
 ATTITUDE MON - AGS
 SUIT FAN - 1
 O2/H2O OTY MON - ASC 2

8
 ENG GMBL - ENABLE
 DES ENG CMD OVRD - OFF
 LDG ANT - DES
 RADAR TEST - OFF
 TEST/MONITOR - ALT XMTR
 SLEW RATE - HI
 RNDZ RADAR - SLEW
 DEAD BAND - MIN
 GYRO TEST - ROLL
 ATTITUDE CONTROL (3) - MODE CONT
 MODE CONTROL - ATT HOLD
 TEMP MON - LDG
 RCS SYS A/B-2 QUAD 1,2,3,4(4) - OFF
 SIDE PANELS - OFF
 FLOOD - ALL
 EXTERIOR LTG - DOCK
 LAMP/TONE TEST - OFF
 X-POINTER SCALE - HI MULT

9
 ACA/4 JET (2) - ENABLE
 TTCA/TRNSL (2) - ENABLE
 AOT - CL, ANGLE 0000°

10
 TTCA (LMP) - JETS (DN)

11
 AUDIO CONT - All Switches Off

12
 VHF A & B XMTR & RCVR - OFF

13
 TRACK MODE - OFF

Basic Date Feb. 24, 1969
 Changed

14

SUIT GAS DIVERTER - PUSH/CABIN
 CABIN REPRESS - CLOSE
 PLSS FILL - CLOSE
 1 & 2 ASC 02 - CLOSE
 SUIT CIRCUIT RELIEF - AUTO
 CABIN GAS RETURN - AUTO
 CO2 CANISTER SEL - PRIM
 WATER SEP SEL - PULL/SEP 2
 (Master Alarm - On)
 SEC EVAP FLOW - CLOSE
 PRIM EVAP FLOW No. 2 - CLOSE
 WATER TANK SELECT - DES
 SUIT TEMP - COLD
 CABIN TEMP - MAX COOL

15

FWD CABIN RELIEF AND DUMP - AUTO

16

CB(11)

Row 1:

A 11 Open Except:
 AC BUS B: NUM LTG - Close
 BUS TIE INV 2&1 - Close
 AC BUS A: BUS TIE INV 2&1 - Close
 AC BUS VOLT - Close

Row 2:

A 11 Open Except:
 FLT DISP: MISSION TIMER - Close
 (Master Alarm May Come On With QUAD TCA)

Row 3:

A 11 Open Except:
 HEATERS: RNDZ RDR STBY - Close
 LDG RDR - Close
 INST: SIG CONDR 1 - Close
 Master Alarm - On With ATCA PGNS

Row 4:

LTG: ANUN/DOCK/COMPNT - Close
 A 11 Open Except:
 SUIT FAN 1 - Close
 ECS GLYCOL PUMP 2&1 - Close
 GLYCOL PUMP AUTO TRNFR - Close
 PGNS: IMU STBY - Close
 (LGC Warning Lt-On When LGC/DSKY Opened)

Basic Date — Feb. 24, 1969
 Changed — Feb. 25, 1969

SYS-72

Row 5: All Open Except:
 EPS: BAT FEED TIE (2) - Close
 CROSS TIE BAL LOADS - Close
 XLUNAR BUS TIE - Close
 DES ECA CONT - Close
 DES ECA - Close
 DC BUS VOLT - Close
 R/D INST A&B (2) - Close
 DFI PWR - ON

17 CB(16)
 Row 1: All Open Except:
 FLT DISP: EVENT TIMER/SE FDAI - Close
 Row 2: All Open Except:
 LTG: FLOOD - Close
 MASTER ALARM - Close
 STAB/CONT: ASA - Close
 (Master Alarm - On With ATCA)
 INST: CWEA - Close
 SIG SENSOR - Close
 PCM/TE - Close
 SIG COND 2 - Close
 Row 3: All Open Except:
 COMM: DISP - Close
 PRIM S-BD XMTR/RCVR - Close
 PMP - Close
 ECS: DISP - Close
 Row 4: HEATERS: S-BD ANT - Close
 EPS: DISP - Close
 DC BUS VOLT - Close
 INV 2 - Close
 DES ECA - Close
 DES ECA CONT - Close
 XLUNAR BUS TIE - Close
 CROSS TIE BAL LOADS - Close
 BAT FEED TIE (2) - Close

Basic Date Feb. 24, 1969
 Changed

18 Install Window Shades

L. v

**FINAL DEACTIVATION

10 min

- 1 Wait Until Dryout Complete (53 min) ~~GLYCOL~~
~~Temp Should Not Go Higher Than 95°)~~
 R/D TLM CAL - ON
 DFI PWR - OFF
 GLYCOL - PUMP 2
 SUIT GAS DIVERTER - PULL/EGRESS
 CABIN TEMP - NORMAL
- 2 S-BAND - PM, OFF, OFF, OFF, OFF, OFF, OFF, HI
 TAPE - OFF
- 3 INV- OFF (Possible Master Alarm - On)
 TIMER CONT - STOP
 DET - RESET/STOP
 ANUN/NUM - DIM
- 4 CB(11)
 Row 1&2: All Open
 Row 3: All Open Except:
 HEATERS: RNDZ RDR/STBY - Close
 LDG RDR - Close
 INST: SIG CONDR 1 - Close
 LDG: ANUN/DOCK/COMPNT - Close
 Row 4: All Open Except:
 (Master Alarm On When Glycol Pump 2
 Opened)
 CB PGNS: IMU STBY - Close
 Row 5: All Open Except:
 EPS: BAT FEED TIE (2) - Close
 CROSS TIE BAL LOADS - Close
 DES ECA CONT - Close
 DES ECA - Close
 DC BUS VOLT - Close
 DFI POWER - OFF
- LM-3 CB(16)
 Row 1: All Open:
 Row 2: All Open Except:
 LTG: FLOOD - Close
 STAB/CONT: ASA - Close
 INST: SIG SENSOR - Close
 SIG CONDR 2 - Close

Basic Date Feb. 24, 1969
 Changed Feb. 25, 1969

Row 3: All Open
 Row 4: All Open Except:
 HEATERS: S-BD ANT - Close
 EPS: DISP - Close
 DC BUS VOLT - Close
 DES ECA - Close
 DES ECA CONT - Close
 BAT FEED TIE (2) - Close

Check BAT & BUS Voltages

BAT 1 _____, BAT 2 _____, BAT 3 _____
 BAT 4 _____, BAT 5 _____, BAT 6 _____
 CDR BUS _____, SE BUS _____
 ED BAT A _____, ED BAT B _____

6 BAT 1, LO VOLTAGE - OFF/RESET tb-bp
 BAT 1, LO VOLTAGE - ON tb-LO
 Repeat For BATS 2,3,4

Check BAT & BUS Voltage & Amps Then
 ED/OFF

7 CB(11) INST: SIG COND 1 - OPEN
 EPS: DES ECA CONT - Open
 DC BUS VOLT - Open
 CB(16) INST: SIG SENSOR - Open
 SIG CONDR 2 - Open
 EPS: DISP - Open
 DC BUS VOLT - Open
 DES ECA CONT - Open
 CROSS TIE BAL LOADS - Close

8 DES H2O - CLOSE
 CSM Position LM PWR - CSM
 FLOOD - OFF (Lights Remain On)
 UTILITY LIGHTS (Both) - OFF
 CSM LM Power _____ (TBD)

9 OVHD CABIN DUMP VALVE - AUTO
 Ingress CSM With LMP SYSTEMS Checklist
 and Secure Hatch

Basic Date Feb. 24, 1969
 Changed _____

L. 3

FIRE/SMOKE IN CABIN (Not In Suit Loop)

1

PRESS REG A&B - EGRESS
 SUIT GAS DIVERTER - EGRESS
 Check POWER/TEMP MON For Excessive
 Current, Remove Power From Affected
 Bus
 Use Fire Extinguisher As Required
 Don Helmets And Gloves
 SUIT FAN - Redundant Fan
 Combustion Products Should Be
 *Considered Toxic. Smoke And *
 *Contaminants Must Be Removed *
 *From Cabin Before Removing *
 Helmets By Purging or Dumping
 *Cabin. *

2

If Fire Persists
 Prepare To Dump Cabin (If Ascent
 Stage Only, Insufficient O₂
 Available For Repress)
 Visually Perform Suit Integrity Check
 CABIN GAS RETURN - EGRESS
 CABIN DUMP (Fwd) - Open, Then Auto
 At 3.2 psia (Verify Suit Press -
 3.6-4.3 psi)
 Cabin Dump Valve Open, Until Cabin
 Press 0 psi

3

When Fire Goes Out
 SUIT GAS DIVERTER - PULL/EGRESS
 SUIT CIRCUIT RELIEF - AUTO
 CABIN GAS RETURN - EGRESS
 CO₂ Canister - MID Position
 PRESS REG B - EGRESS
 PRESS REG A - DIRECT O₂ (Until ARS
 Clear; Suit Press Will Increase To
 About 5.8 psia)
 CO₂ Canister Sel - PRIM

Basic Date Feb. 24, 1969
 Changed Feb. 25, 1969

EMER-2

FIRE/SMOKE IN SUIT LOOP (MAY BE IN CABIN ALSO)

- 1
SUIT ISOL (Both) - SUIT DISC
SUIT FAN - OFF
PRESS REG A&B - CLOSE
Combustion Products Should Be Con-
sidered Toxic. Smoke And Contam-
*inants Must Be Removed From Cabin *
Before Removing Helmets By Purging
*or Dumping Cabin *
Remove Helmet & Gloves
SUIT CIRCUIT RELIEF - CLOSED
CABIN GAS RETURN - EGRESS
SUIT GAS DIVERter - PULL/EGRESS
- 2
Isolate SUIT LOOP Electrically
CB(11)ECS: SUIT FAN 1 - Open
CB(16)ECS: SUIT FLOW CONT - Open
SUIT FAN 2 - Open
SUIT FAN Δ P - Open
DIVERter VLV - Open
CO2 SENSOR - Open
CABIN REPRESS - MANUAL As Necessary
To Maintain Cabin Pressure And
Replenish O2
- 3
When Fire Goes Out
If Cabin Contaminated - Purge As Necessary
CABIN REPRESS - MANUAL
CABIN DUMP VALVE - AUTO
If Ascent Stage Only, Closely Monitor
O2 Supply

Basic Data
Changed

WARNING LIGHTS LM-3**AGS**

1. AEA Test Mode Discrete Signals Fail Condition
2. ASA Heater Fails On Causing A Temp Sensor To Open The +12 vdc Supply
3. Power Supplies Out Of Limits

1. Switch To PGNCS Control
2. Perform AGS Self Test 412R (+1 Test Passed)

ASC PRESS

1. He TANK < 2775 psia (Inhibited after Staging)
2. FUEL, OXID TANK < 120 psia

1. Cross-Check PRPLNT TEMP/PRESS MON - ASC
2. Shut Down APS When Press < 120 psi
3. Close ASCENT He REGS 1&2 And Check For Leak

CABIN

1. Cabin Pressure < 3.7-4.45 psia (Light Disabled In Egress Position)

1. Cross-Check Cabin Press, Suit Press, & Cuff Gages
2. Close CABIN DUMP VLVS, Check PRESS REGS A&B - CABIN, CAB IN REPRESS - AUTO
3. Don Helmet & Gloves, Then Turn CABIN REPRESS - CLOSE And PRESS REGS A&B - EGRESS, CABIN GAS RETURN - EGRESS, SUIT GAS DIVERT - PULL/EGRESS To Troubleshoot

CES AC

1. CES AC Pwr Supplies Out Of Tolerance

.. GUID CONT - PGNCS
(No Rate Damping Or Attitude Control Or No Manual SPS Throttling In AGS)

EMER-4

CES DC

1. CES DC Pwr Supplies Out Of Tolerance

1. GUID CONT - PGNCs

(No DPS Throttling In PGNS Or AGS, AGS Direct Still Available, DPS May Go To 100%, Deadband Inoperative)

DC BUS

1. Either Or Both DC Buses < 26.5 V

1. Check For BATTERY CAUTION LT, BATT Or DC BUS Component Lights, Check All BAT & BUS Voltages, And Reset All BAT tb's.
2. If No Other Lights - Separate Buses By Opening CROSS TIE BAL LOADS cb And Observing PWR/TEMP MON. If Volts And Amps Not Normal Power Down Low Bus, Open All cb's, Check For Bus Or Feeder Short By Observing DC BUS FAULT LT (If SE BUS) And PWR/TEMP MON (If CDR's BUS)
- 3a. If DC BUS FAULT LT ON (And Unstaged) Indicates ECI Or DFR OPEN - Use BAT 6 NORMAL CDR FEED-ON.
- 3b. If DC BUS FAULT LT ON (And Staged) Power Down Bus, Open All cb's (Except BAT FEED TIE(2) And DC BUS VOLT). If DC BUS FAULT LT Still On Indicates BAT 6 Lost, Reconfigure With BAT 5 Backup Feed. If DC BUS FAULT LT OUT - Indicates Failed ECA.
4. If BATTERY CAUTION LT And DC BUS LT ON - CDR Bus Power Lost - Check BAT tb - If Only One BAT bp (Unstaged Only) Remove Other BAT And Reset The bp BAT. If Both BATS bp (Or 1 Bat bp Staged) Check For Bus Or Feeder Short By Opening All cb's (Except DC BUS Volts) And Observing DC BUS FAULT LT - If LT OFF, CDR BUS FEED SHORT.
5. (For LMP BUS POWER FAILURE, C/W PWR CAUTION LT WILL BE ON) And Flood And Integral Lights Will Be Off - See Procedures Under C/W PWR Procedures

Basic Date _____

Changed _____

EMER-5

DES QTY

1. Burn Time < 2 Min @ 25%
2. FUEL Or OXID < 6%
(Only ARMED When DPS's ON)

1. Cross-Check PROP QTY
2. Shutdown DPS Before Depletion

DES REG

1. He Downstream Of REGS < 220 Or > 260psi

1. Close REG 1 & Open REG 2
2. If Press Decay Continues Below 140 psi Shutdown DPS
3. Close REG 2

ISS

1. IMU Fails, ICDU Fails, Or PIPA Fails
During Thrust

1. Switch To AGS Control
2. Perform MANUAL Engine Shutdown

LGC

1. LGC Prime Pwr, Scalar, Or Counter Fails

1. Switch To AGS Guidance
2. Perform MANUAL Engine Shutdown

RCS A REG

1. He REG PRESSURE < 165 Or > 205 psi

1. Cross-Check PRPLNT FUEL MANF And OXID MANF Pressures
2. If Press Below 140 Or Above 205 psi, Turn Off
SYS A MAIN SOV And Use CROSSFEED

RCS B REG

1. He REG PRESSURE < 165 Or > 205 psi

1. Cross-Check PRPLNT FUEL MANF And OXID MANF Pressures
2. If Press Below 140 Or Above 205 psi, Turn Off
SYS B MAIN SOV And Use CROSSFEED

EMER-6

RCS TCA

1. One Or More Thrusters Fail Off
2. Colinear Thrusters Fire Simultaneously
3. Detects Thruster Failed On If Rate Feedback In Effect

1. If Only One Red Flag, Close QUAD VALVE
If More Than One Red Flag In Same SYS, Open
MAIN SOV or ASC FEED In Affected SYS A or B
2. Complete Translation If Translating
3. If Thruster Firing Continuously, Turn Off QUAD
VALVE With Red Flag (This Would Fix Stuck-On
Thruster)
4. If Not Continuously Firing Recycle All RCS
VALVES To Normal Position - Use TCA cb's And
Rate Command Attitude Control To Troubleshoot

SUIT/FAN

1. Suit Pressure < 3.12 psi
2. Suit Fan #2 Fails While It Is Selected

Cabin Depressurized

- 1a. Cross-Check With SUIT And CABIN PRESS, And With
PGA CUFF GAGE
- 1b. Check PRESS REG A&B - EGRESS. If Press Still
Too Low Set PRESS REG A&B - DIRECT O2 Inter-
mittently To Hold Suit Press Up Until Cabin
Can Be Repressurized
- 1c. If Suit Isol Valves Closed And Suit Integrity
OK, Pull CB(16) ECS: Suit Flow Cont - Open &
SUIT ISOL VLVS To SUIT FLOW

Cabin Pressurized

- 2a. Cross-Check SUIT And CABIN PRESS And CUFF GAGE
- 2b. Check Suit Flow If SUIT FAN 2 Selected
- 2c. If Suit Isol Valves Closed - Pull CB(16) ECS:
SUIT FLOW Cont - Open & Suit ISOL vlvs To
SUIT FLOW

Basic Date _____

Changed _____

CAUTION LIGHTS LM-3

ASC HI REG 1. He Manifold Press > 220 psi

Continue Engine Burn And Close Both ASC He REG 1 & 2

2. Cross Check ASC PRESS
3. When He Press Drops Below 220, Open REG 1 - If Light Comes On Again Close REG 1 And Open REG 2

ASC QTY 1. When About 10 Sec Of Propellant Burn Time Remains

1. Stop ASC Fuel And ASC Oxid Feed To RCS
2. Cross-Check With ASC He Pressure
3. Shutdown ASC Engine

BATTERY 1. Bat Reverse Current > 10 Amp For 4-6 Sec
 2. Bat Temp > 145°F
 3. (Bat Overcurrent Trip Will Result In Additional Loss)

1. Check All Bat tb's. If bp, Try Resetting Bat OFF/RESET Then On
2. Check All Bat Positions And Note On Which Bat The Bat Fault Component Lt Illuminates
3. Remove Faulty Bat From Bus

C/W PWR 1. No C/W Warning Pwr
 2. Illuminates With Loss Of Entire LMP Bus

- a. No Other C/W Lights (C/W Failure) - RESET CWEA cb
- a. DC Bus Fault Lt - ON, Flood And Integral Lighting Off (LMP Bus Failure)
- 2b. LMP AUDIO CONT - BU, Pwr Up INV 2, And Open All cb's On LMP Panel (16)
- 2c. If DC Bus Fault Lt Goes Off, Bat To Bus Feeder Open. Set BAT 5 NORMAL SE FEED-ON, Close All cb(16)

EMER-8

- 2d. If DC Bus Fault Lt Stays On, Take Each Bat Off Line In Sequence And Monitor The DC Bus Fault Lt. When Lt Goes Out It Signifies Faulty Bat Removed. Reconfigure With ASCENT BAT 5 And/Or CROSS TIE BUS cb's.
- 2e. If DC Bus Fault Lt Stays On With Both Bats Removed, Open Both BAT FEED TIE cb's. If DC Bus Fault Lt Goes Out It Indicates A Feeder Short. Reconfigure With Bat 5. If Lt On Indicates LMP Bus Short.

ECS

1. Suit Fan $\Delta P < 6''$ H2O
2. CO2 Partial Pressure > 7.6 mm Hg
3. H2O Sep Speed < 800 rpm
4. Glycol Pump $\Delta P < 3$ psid

1. Cross-Check Component Lights And Select Redundant Component
2. If No Component Lights On, CWEA Failure

ED RELAYS

1. Light Illuminates If Contacts Of Master Arm Relay or Staging Sequence Relays Fail Closed (Both Stage System Lights Should Be On When Master Arm - On, And Off When Master Arm - Off)

1. Do Not Turn On Master Arm Switch
2. Attempt To Reset Stage Relay - If Unable Pull Appropriate ED Logic Pwr

GDA/RCCA

1. Light On If There Is Discrepancy Between Gimbal Drive Signal And Gimbal Response During DPS Burn

1. Monitor RCS Duty Cycle And Attitude Rates. If Increasing, Set ENG GIMBAL - OFF
2. If RCS Duty Cycle Still Excessive, Shutdown The DPS

Basic Date _____
Changed _____

EMER-9

GLYCOL

1. Glycol Temp $> 50^{\circ}\text{F}$
2. Glycol Remaining In Accumulator $< 10\%$

Cross-Check Glycol Temp And Press, CABIN And SUIT TEMPS, And H2O QTY

- 2a. If Glycol Temp $> 50^{\circ}\text{F}$ Observe Rate Of Glycol Temp Increase. If Temp Steady Suspect Thermal Overload Or INST/CWEA Failure
- 2b. If Increasing, Recycle PRIM EVAP FLOW - OPEN
- 2c. If Glycol Temp Continues To Increase, Close PRIM EVAP FLOW And Set PRIM EVAP FLOW #2 OPEN
- 2d. If Glycol Temp Continues To Increase, Activate Secondary Loop By WATER TANK SEL-SEC, GLYCOL INST(SEC), cb GLYCOL PUMP SEC - Close, SEC EVAP FLOW OPEN Then Shut Down Primary Loop
3. If Glycol Temp $< 50^{\circ}\text{F}$ Suspect Low Glycol Quantity, Monitor Glycol Temp, Press Gages

HEATER

1. RR - Temp $< -54^{\circ}\text{F}$ or $> 148^{\circ}\text{F}$
2. LR Heater (C&W Not Operable In LM-3)
3. RCS- Temp $< 119^{\circ}\text{F}$ or $> 190^{\circ}\text{F}$
4. S-Bd-Temp $< -64^{\circ}\text{F}$ or $> 153^{\circ}\text{F}$

- 1a. RR Too Hot - Check For Heat Soak, Open Both RNDZ HTR cb's (If Still Too Hot Open CB AC BUS A: RNDZ RDR, Then CB PGNS: RNDZ RDR When RR Not Needed)
- 1b. If Too Cold - Power Up RNDZ RDR
- 2a. LR Too Hot - Open LDG RDR HTR CB & CB PGNS: LDG RDR When LR Not Needed (Note: When CB PGNS Is Closed, Relay Open Line Between CB HTR: LDR RDR & LR Heaters)
- 2b. LR Too Cold - Power Up LDG RDR
- 3a. RCS Too Hot - QUAD A/B-1 & 2 - Open
- 3b. RCS Too Cold - QUAD A/B-1(or 2) - Closed, RCS SYSTEMS A/B-2 Sw - MANUAL
- 4a. S-BD Too Cold - CB COMM: S-BD ANT - Close
- 4b. S-BD Too Hot - CB COMM: S-BD ANT - Open & CB HTR: S-BD-Open (When S-BD not Needed)

Basic Date
Changed

EMER-10

INVERTER

1. AC Voltage < 112 Volts
2. Frequency < 398 cps > 402 cps

1. Check All AC Bus A/B: BUS TIE INV 1&2 cb's (4)
2. If Some cb's Open Determine If INV 1 Feeder Short Or AC Bus A Or Bus B Short By Pulling BUS TIE INV 1 cb's And Closing Bus Tie INV 2 cb Then Monitor cb's
- 3a. If All cb's Remain Closed Cross-Check With Pwr/Temp Mon
- 3b. If Volts In Green - Determine If INV 2 Or CWEA Failure By Powering Up INV 1
- 3c. If Volts Not In Green - Determine If INV 2 Or INV 2 Feeder Short By Powering Up INV 1 - If Any AC BUS A/B: BUS TIE INV cb2 Open; Indicates INV 2 Feeder Short
4. If All cb's Do Not Remain Closed - Check For AC Bus Short Or INV 1 Feeder Short By Opening Both AC BUS A/B BUS TIE INV 1 cb's And Closing Both AC BUS A/B BUS TIE INV 2 cb's.

LDG RDR

If Light Comes On It Is A Failure Of The LR Data Good Circuit Or CWEA

02 QTY

1. Descent Qty < 5%
2. Either Ascent 02 Tank Qty < 80% Prior To Staging
3. Ascent 02 Tank #1 Qty < 10% When Staged

1. Cross-Check 02 Qty Gage and Cabin Press
- 2a. If Cabin Press Normal and 02 Leak Outside Cabin Or Cabin Leak. Close Dump Vlv's And Put Cabin Repress - Close And PRESS REGS A&B - EGRESS
- 2b. If Cabin Press High - Check PRESS REGS A&B, CABIN REPRESS, PLSS VLVS For Possible Failures
3. If Descent 02 Lost, Go To ASC 1 And Close CABIN REPRESS, SUIT GAS DIVERter - PULL/EGRESS And CABIN GAS RETURN - EGRESS

Basic Date _____
Changed _____

PRE AMPS

1. Lt On If Either RCS Bias Pwr Supply To The Prim Preamped & Jet Drivers Out Of Limit (One From cb ATCA And One From cb ATCA [PGNS])

1. If Both Bias Voltages Out Of Tolerance There May Be Sporadic RCS Firings

RCS

1. He Tank Press < 1700 psia

1. Cross-Check RCS He PRESS, [PRPLNT, And FUEL MANF And OXID MANF PRES] (The Bracketed Positions Are Useful Only If There Has Been a Propellant Leak Followed By a Bladder Rupture). Cross-Check RCS Qty (Qty Gage Uses He Press)
2. Use Both SYS When FUEL or OXID MANF press 140 psia, Close Bad Systems MAIN SOV For RCS Burns, Use ASC FEED Or CROSSFEED From Good RCS (RCS Fuel And Oxid In Failed System Unusable)

RNDZ RDR

1. When RR Is In Auto Track Mode And Loses CSM Lock-On (NO TRACK LT Should Be On Also)

1. Determine If RR Is Tracking CSM
2. Check For Momentary Data Loss By Selecting MODE CONT - SLEW Then AUTO TRACK
3. Check AGC And XMTR PWR. If Low, Cycle cb PGNS-RNDZ RDR To Eliminate Chrona
4. Attempt Reacquisition With P20 or Visually With AUTO TRACK Or With AGS 400 + 2 And AUTO TRACK Check CSM Transponder

WATER QTY

1. Descent H2O Qty $< 16\%$
2. Either Ascnet H2O Tank Qty $< 95\%$ Unstaged
3. H2O In Ascent Tank #1 And #2 Differ By $> 15\%$

EMER-12

1. Cross-Check With H2O Gage
2. Verify H2O TANK SELECT In Proper Position
3. Verify SEC EVAP FLOW And PRI EVAP FLOW #2 - CLOSED
4. Monitor Glycol Temp And When $> 50^{\circ}$ Select ASC H2O Tank

COMPONENT CAUTION LIGHTS

DC Bus Fault

Illuminates When Voltages Between Buses Differ;
Fully Bright When Voltages Differ By > 18 Volts.
When All cb's Are Opened On A Suspect BUS, Then
BUS CROSS - TIED To the Other Bus. Light On Will
Indicate A Bus That Exists.

Battery Fault

Illuminates When That Specific Battery Temp $> 145^{\circ}\text{F}$,
Reverse Current > 10 Amps For 4-6 Sec, Overcurrent
Has Disconnect Bat $> 150-200$ Amp.

Range/Range Rate Power Failure Light

Loss Of:

1. Range
2. Range Rate
3. PCM/TE
4. AC Power To Meter

Suit Isolation Valves

Closes At 3.11 psia. Cabin Repress Will Be
Activated At The Same Time. (If Enabled)

Cabin Repress

Will Activate At 3.70-4.45 And Close At 4.40-5.0.

Basic Date _____
Changed _____

UPDATE PADS, PHOTO LOGS, CREW COMMENTS

Basic Date Jan. 1, 1969
Changed

LM-3

LM P27 UPDATE

V		V		V		PURP
INDEX		INDEX		INDEX		
						01 306
						02 307
						03 310
						04 311
						05 312
						06 313
						07 314
						10 315
						11 316
						12 317
						13 320
						14 321
						15 322
						16 323
						17 324
						20 325
						21 326
						22 327
						23 330
						24 331
:	:	:	:	:	:	T
:	:	:	:	:	:	$\phi (+N)$
:	:	:	:	:	:	$\lambda (+E)$
:	:	:	:	:	:	NAV
+	.	+	.	+	.	H CHECK

REMARKS